

TD SECURITIES (USA) LLC

April 7, 2026

Price: \$22.61 (04/6/2026)

Price Target: NA
BUY (1)
ESG SCORE: 58/100
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Key Data

Symbol	NASDAQ: MPLT
Beta:	(2.79)
52-Week Range:	\$22.61-\$12.81
Market Cap:	\$959.5MM
Net Debt (MM):	\$(299.3)
Cash/Share:	\$(3.15)
Dil. Shares Out (MM):	44.0
Enterprise Value (MM):	\$696.1
BV/Share:	\$10.41
Dividend:	\$0.00
Yield:	0.0%

FY (Dec)	2025A	2026E	2027E
EPS			
Q1	-	\$(1.08)	-
Q2	-	\$(1.15)	-
Q3	\$(37.18)	\$(1.16)	-
Q4	\$(2.47)	\$(1.10)	-
Year	\$(18.56)	\$(4.50)	\$(5.15)
Consensus EPS	\$(18.56)	\$(4.06)	\$(4.10)
Consensus source: FactSet			

Revenue (MM)

Year	\$0.0	\$0.0	\$0.0
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INITIATING COVERAGE

Initiating At Buy: Charting A Course Toward Improved CNS Therapeutics

THE TD COWEN INSIGHT

Maplight is advancing candidates for the treatment of CNS and neuropsych conditions. Lead asset ML-007C-MA looks to improve upon Cobenfy's profile in the treatment of schizophrenia with Ph. II data set for Q3. A Ph. II study in Alzheimer's psychosis would expand the therapy's opportunity. ML-004 is in a Ph. II study for ASD (Q3:26 data). We see shares as undervalued for schizophrenia alone. Buy.

Maplight Looks To Improve Upon The Muscarinic Profile For Neuropsychiatric Conditions

Maplight Therapeutics is a biotechnology company focused on the development of therapies for the treatment of CNS and neuropsychiatric conditions. Lead asset ML-007C-MA is a M1/M4 agonist in Ph. II development designed to have an improved targeting and potency profile vs. BMS's Cobenfy. While Cobenfy has been a welcomed addition to the schizophrenia treatment paradigm, the agent does have problematic GI side effects and a more challenging administration profile that could limit broader use. Maplight believes ML-007C-MA's profile will be able to deliver on measures of disease activity while limiting peripheral cholinergic side effects, and provide benefit on additional measures of disease (cognition etc.) with a more convenient dosing profile (e.g. QD dosing and/or lack of need for fasting). Ph. II data in the indication are expected in Q3:26.

While proof of differentiation vs. Cobenfy will be key, our KOLs note there is much room to improve on the therapy's profile, and we model peak U.S. ML-007C-MA sales of \$1.3B, and an NPV of the schizophrenia opportunity alone results in a value of \$35/share. Thus, we see shares as undervalued for the potential of ML-007C-MA alone. Outcomes from patients treated with agents targeting the muscarinic pathway have led MPLT and others to pursue development in Alzheimer's psychosis. There are currently no approved agents to treat psychosis in this patient population, and thus this market represents meaningful untapped potential for the class. Given the lack of established data in the indication, we see revenues in ADP as upside to our model. Ph. II data are expected in H2:27.

Pending results in these two indications, we see potential for MPLT to study ML-007C-MA or other muscarinic agents in a broader development program given the potential utility of the mechanism across Alzheimer's agitation, autism spectrum disorder, bipolar disorder and others, which represent potential upside to our model. While targeting the muscarinic pathway has yielded successes and failures, we see meaningful potential in the mechanism and initiate at Buy.

Optimized Zolmitriptan Formulation Hopes To Improve Social Dynamics Of Autism Spectrum Disorder

MPLT is advancing ML-004, a 5-HT1B therapeutic to improve social communication challenges in patients with ASD. There are few agents approved for the treatment of ASD and they only target the irritability components of the disease. Pre-clinical data suggest that the serotonin pathway is implicated in the social aspects of ASD, with the potential to improve these symptoms through agonism at the serotonin receptor. Ph. II data are expected in Q3. Earlier in the pipeline MPLT is advancing a M4 antagonist for Parkinson's and a GPR52 PAM for hyperactivity/impulsivity. While much development remains, MPLT has several shots on goal and a modest valuation. The company is funded through initial catalysts from the pipeline.

AT A GLANCE

Our Investment Thesis

Maplight Therapeutics' lead asset is ML-007C-MA, a M1/M4 agonist combined with a peripherally acting anti-cholinergic agent. While BMS's Cobenfy has delivered an efficacious new agent to the schizophrenia treatment paradigm, Cobenfy has tolerability concerns and a more challenging administration profile. MPLT has developed ML-007C-MA to offer potentially QD dosing and/or improved administration through not requiring fasting while also optimizing targeting for efficacy and tolerability. A Ph. II study in schizophrenia will yield data in Q3:26, and we see \$1.3B in U.S. peak revenues, if successfully developed. A Ph. II trial in Alzheimer's psychosis is also ongoing with data expected in H2:27. MLPT's other clinical candidate is ML-004, a 5-HT1B agonist to improve social communication challenges in ASD. A Ph. II study will yield data in Q3:26. We see shares as undervalued for schizophrenia alone, with additional upside possible through indication expansion or the earlier pipeline.

Base Case Assumptions

- ML-007C-MA is successfully developed for the treatment of schizophrenia and achieves blockbuster status.
- ML-007C-MA advances in ADP and/or other CNS/neuropsychiatric conditions
- Maplight is able to secure funding to advance its portfolio

Upside Scenario

- ML-007C-MA is successfully developed for both schizophrenia and Alzheimer's disease psychosis
- ML-004 is successfully developed for the treatment of ASD symptoms
- Maplight's early pipeline efforts result in successful clinical candidates

Forthcoming Catalysts

- Ph. II data of ML-007C-MA in schizophrenia (Q3:26)
- Ph. II data of ML-004 in autism spectrum disorder (Q3:26)
- Ph. II data of ML-007C-MA in Alzheimer's psychosis (H2:27)

Downside Scenario

- ML-007C-MA is not successfully developed for the treatment of schizophrenia
- ML-007C-MA encounters tolerability concerns or lacks a differentiated profile vs. Cobenfy
- Maplight is unable to secure funding to advance its portfolio

Price Performance



Source: Bloomberg

Company Description

Maplight Therapeutics is a clinical-stage biotechnology company focused on the development of agents to treat CNS and neuropsychiatric conditions. Lead asset ML-007C-MA is a M1/M4 agonist combined with a peripherally acting anti-cholinergic agent in development for schizophrenia and Alzheimer's disease psychosis. The company is hopeful the therapy will be able to deliver an improved profile vs. commercial treatments targeting the muscarinic pathway and Ph. II data are expected in Q3:26 and H2:27, respectively. Also in Ph. II is ML-004, a targeted serotonin receptor agonist (and formulation of zolmitriptan) being developed to address the social challenges associated with ASD. Data are expected in Q3:26. Earlier in the pipeline, MPLT has a M4 antagonist for Parkinson's disease, and a GPR52 PAM targeting the indirect pathway for indications of hyperactivity/impulsivity. The companies muscarinic agents may also have applicability outside the indications currently in clinical development.

Analyst Top Picks

	Ticker	Price (04/6/2026)	Price Target	Rating
United Therapeutics	UTHR	\$559.65	\$660.00	Buy
Axsome Therapeutics	AXSM	\$169.74	\$215.00	Buy
Entrada Therapeutics	TRDA	\$13.88	NA	Buy

Investment Thesis

Maplight Therapeutics is a clinical-stage biotechnology company focused on the development of agents to treat CNS and neuropsychiatric conditions. Lead asset ML-007C-MA is a M1/M4 agonist combined with a peripherally acting anti-cholinergic agent in development for schizophrenia and Alzheimer's psychosis. While BMS' (through the acquisition of Karuna) Cobenfy has been a welcomed addition to the schizophrenia treatment paradigm, the agent does have problematic GI side effects and a more challenging administration profile that could limit broader use, per our KOL discussions. Maplight believes ML-007C-MA's profile will be able to deliver on measures of disease activity while limiting peripheral cholinergic side effects, and provide benefit on additional measures of disease (cognition etc.) with a more convenient dosing profile (e.g. QD dosing and/or lack of need for fasting). A Ph. II study is expected to complete enrollment in April, with data expected in Q3:26. While a differentiated profile vs. Cobenfy will be key, and there are other muscarinic-based agents in development from companies including NBIX, ABBV, and NMRA, should ML-007C-MA be approved with a compelling profile, our KOLs see much room for improvement on the current SOC options. We model peak U.S. sales of \$1.3B in 2040 for the therapy. This along with an assumed ex-U.S. partnership results in an NPV-value of \$35/share.

The same therapy is also in Ph. II development for the treatment of Alzheimer's disease psychosis. While our KOLs indicate that the unmet need in this population is high given there are not formal FDA-approved agents to treat the indication, and the side effects associated with off-label atypical anti-psychotic agents are poor, we currently leave success in this indication as upside to our estimates as we see initial results from the class (BMS's Cobenfy will have data this year, and MPLT's Ph. II will yield data in H2:27). The muscarinic pathway could be implicated in additional neuropsychiatric indications including Alzheimer's agitation, autism spectrum disorder, bipolar disorder and others. Pending data from these initial studies, MPLT could expand development into these indications either with ML-007, or next-generation candidate ML-055.

Outside of the company's muscarinic pathway-targeted agents, MPLT is advancing ML-004, a 5-HT1B therapeutic to improve social communication challenges in patients with ASD. There are few agents approved for the treatment of ASD and they only target the irritability components of the disease. Our KOLs note that essentially any statistically significant benefit that would drive approval would lead to broad use of a therapeutic given the high unmet need. Ph. II data are expected in Q3, and ML-004 remains upside to our modeled estimates given the challenging development landscape. Earlier in the pipeline MPLT is advancing a M4 antagonist for Parkinson's and a GPR52 PAM for hyperactivity/impulsivity. While much development remains, MPLT has several shots on goal and a modest valuation. We initiate at Buy.

Maplight Therapeutics Upcoming Milestones

Milestone	Timing
Data from Ph. II ZEPHYR study of ML-007C-MA in schizophrenia	Q3:26
Data from Ph. II IRIS study of ML-004 in autism spectrum disorder	Q3:26
Complete IND-enabling studies for ML-021 in the treatment of Parkinson's disease	H2:26
Data from Ph. II VISTA study of ML-007C-MA in Alzheimer's disease psychosis	H2:27

Source: TD Cowen

Maplight Therapeutics Quarterly P&L Model (\$MM)

	H1:25A	Q3:25A	Q4:25A	2025A	Q1:26E	Q2:26E	Q3:26E	Q4:26E	2026E
ML-007C-MA Revenue	-	-	-	-	-	-	-	-	-
Total Revenue	\$0.0	\$0.0	\$0.0	\$0.0	\$0.0	\$0.0	\$0.0	\$0.0	\$0.0
COGS	-	-	-	-	-	-	-	-	-
R&D	46.6	27.1	64.6	138.4	35.0	37.5	40.0	40.5	153.0
SG&A	7.6	4.4	18.8	30.7	11.0	11.5	12.0	12.5	47.0
Other	-	-	-	-	-	-	-	-	-
Operating Expenses	\$54.2	\$31.5	\$83.4	\$169.1	\$46.0	\$49.0	\$52.0	\$53.0	\$200.0
Other Income, net	-	-	-	-	-	-	-	-	-
Operating Income / (Loss)	(54.2)	(31.5)	(83.4)	(169.1)	(46.0)	(49.0)	(52.0)	(53.0)	(200.0)
Interest Income (Expense)	1.5	1.5	2.5	5.5	-	-	-	-	-
Other Non-Operating Income, net	0.5	0.5	1.4	2.4	-	-	-	-	-
Pretax net income	(52.2)	(29.4)	(79.5)	(161.2)	(46.0)	(49.0)	(52.0)	(53.0)	(200.0)
Taxes	-	-	-	-	-	-	-	-	-
GAAP Net Income	(52.2)	(29.4)	(79.5)	(161.2)	(46.0)	(49.0)	(52.0)	(53.0)	(200.0)
GAAP EPS	(\$68.46)	(\$37.18)	(\$2.47)	(\$18.56)	(\$1.08)	(\$1.15)	(\$1.16)	(\$1.10)	(\$4.50)
Diluted Shares Outstanding (MM)	0.8	0.8	32.1	8.7	42.4	42.4	45.0	48.0	44.5

Source: TD Cowen

Maplight Therapeutics P&L Annual Model (\$MM)

	2025A	2026E	2027E	2028E	2029E	2030E	2031E
ML-007C-MA Revenue	-	-	-	-	35.7	174.7	390.2
Total Revenue	\$0.0	\$0.0	\$0.0	\$0.0	\$35.7	\$174.7	\$390.2
COGS	-	-	-	-	2.9	14.0	32.0
R&D	138.4	153.0	225.0	250.0	165.0	125.0	75.0
SG&A	30.7	47.0	60.0	65.0	125.0	250.0	295.0
Operating Expenses	\$169.1	\$200.0	\$285.0	\$315.0	\$292.9	\$389.0	\$402.0
Operating Income / (Loss)	(169.1)	(200.0)	(285.0)	(315.0)	(257.1)	(214.3)	(11.8)
Other Non-Operating Income, net	2.4	-	-	-	-	-	-
Pretax net income	(161.2)	(200.0)	(285.0)	(315.0)	(257.1)	(214.3)	(11.8)
Taxes	-	-	-	-	-	-	-
GAAP Net Income	(161.2)	(200.0)	(285.0)	(315.0)	(257.1)	(214.3)	(11.8)
GAAP EPS	(\$18.56)	(\$4.50)	(\$5.15)	(\$5.45)	(\$4.45)	(\$3.65)	(\$0.20)
Diluted Shares Outstanding (MM)	8.7	44.5	55.3	57.8	57.8	58.7	59.0

Source: TD Cowen

Maplight Is Advancing A Pipeline Of Candidates Across Neuropsychiatric And CNS Indications

Maplight Therapeutics is a biotechnology company focused on the development of novel agents for the treatment of neuropsychiatric and neurodegenerative diseases. The company's lead asset is ML-007C-MA (a novel M1/M4 agonist in combination with peripheral antagonist). While the agent shares a similar mechanism with BMS's Cobenfy, Maplight is hoping that optimized agonist and antagonist exposures, with strong activation of both M1 and M4, can help improve broad symptom control, limit AEs seen with the class, and potentially allow for an improved dosing profile.

The agent is currently in a Ph. II trial investigating two different dosing schemes vs. placebo monitoring the change from baseline on the PANSS score at week 5, with data expected in Q3:26. ML-007C-MA is also in development for the treatment of Alzheimer's disease psychosis with primary endpoint data expected in H2:27. As we have previously outlined ([in our neuropsych landscape analysis](#)), placebo response rates in neuropsychiatry trials have been variable. Management believes it is taking several steps to help ensure optimal trial conduct across both indications. Maplight also unveiled a pre-clinical candidate, ML-055, another M1/M4 agonist that management indicates could be more potent than ML-007C-MA and potentially amenable to formulation as a long-acting injectible. The team expects to announce a clinical candidate to advance into IND-enabling studies this year.

Behind the lead muscarinic agent, Maplight is advancing ML-004, a 5-HT_{1B/1D} agonist for the treatment of autism spectrum disorder, currently in a Ph. II study with topline data expected in Q3:26. A M4-specific antagonist agent, ML-021 is currently in pre-clinical studies for the treatment of Parkinson's disease, and a GPR52 PAM, ML-009, rounds out the pre-clinical pipeline with a focus on hyperactivity and impulsivity indications.

Maplight Therapeutics Pipeline

Program	Circuit	Indications	Preclinical	Phase 1	Phase 2	Phase 3	Anticipated Milestones
ML-007C-MA M ₁ /M ₄ agonist co-formulated with PAC	Direct and Indirect Pathways	Schizophrenia	ZEPHYR				Topline results in Q3 2026
		Alzheimer's Disease Psychosis	VISTA				Topline results in 2H 2027
ML-004 5-HT _{1B/1D} agonist	Dorsal Raphe to Nucleus Accumbens	Autism Spectrum Disorder Sociability/Irritability	IRIS				Topline results in Q3 2026
ML-009 GPR52 PAM	Indirect Pathway	Hyperactivity/Impulsivity					Complete IND-enabling studies in 2027
ML-055 Next-Gen M ₁ /M ₄ agonist	Direct and Indirect Pathways	Neuropsychiatric Disorders					Nominate preclinical candidate in 2026
ML-021 M ₄ antagonist	Direct Pathway	Parkinson's Disease					Finalize preclinical candidate in 2027

Source: Maplight Therapeutics

Schizophrenia: Antipsychotics Market Cornered By Atypicals

Schizophrenia is categorized as a psychosis and involves a range of cognitive, behavioral, and emotional symptoms. The DSM-5 (Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition) is typically used to evaluate and diagnose patients. In order to be diagnosed with schizophrenia, a patient must experience two or more of the following symptoms for at least one month: (1) delusions, (2) hallucinations, (3) disorganized speech, (4) grossly disorganized or catatonic behavior, and (5) negative symptoms (i.e. diminished emotional expression). Furthermore, at least one of the symptoms a patient experiences must include (1) delusions, (2) hallucinations, or (3) disorganized speech. Psychotic features of schizophrenia typically emerge when

individuals are aged in the mid-teens to mid-thirties, with the peak age of onset of the first psychotic episode in the early to mid-twenties for males and late twenties for females.

An estimated 20-25MM people worldwide have schizophrenia, with an estimated 3.5MM in the US (or roughly 1% of the population), though our consultants indicate that misdiagnosis of bipolar patients in the past may have led to an overestimation in historical schizophrenia prevalence numbers.

Schizophrenia is generally thought to arise from imbalances in a number of neurotransmitters in the brain, which include dopamine, serotonin, glutamate, and others. Per our KOL consultant, roughly 1/3 of schizophrenia patients will have “mild” symptoms which will enable them to continue to maintain jobs and live somewhat customary lives, while another 1/3 will have serious illness which impairs the patient’s ability to work and maintain typical relationships. Our consultants indicate patients typically receive a diagnosis of schizophrenia approximately 5-8 years after the onset of initial symptoms, though virtually every patient receives an accurate diagnosis at some point in their lives as the condition has a distinctive set of characteristics which grow more obvious as the disease manifests itself. Overall, our KOL consultant indicates that roughly 20-30% of schizophrenia patients remain untreated, in part due to general patient non-compliance and in part due to lack of access to adequate health care options.

Pharmacological management of schizophrenia patients typically includes one or more antipsychotic agents which target dopamine receptors in the brain and primarily address the positive symptoms of the disease (e.g. hallucinations, delusions), though schizophrenia can also be characterized by negative symptoms (e.g. lack of motivation and emotional expression) and cognitive symptoms (e.g. concentration, attention, and memory) for which there are no specifically approved agents. The schizophrenia treatment armamentarium consists primarily of older, “typical” antipsychotic agents (e.g. Haldol, Prolixin), which act mainly as dopamine receptor antagonists in order to reduce the levels of dopamine in patients with schizophrenia, and newer, atypical antipsychotic agents (e.g. Zyprexa, Risperdal), which act as partial dopamine receptor antagonists, but also regulate the levels of serotonin in patients via action at the 5HT_{2A} receptor in particular. Due to the lower affinity of atypical agents for the D₂ dopamine receptor, these products have lower rates of extrapyramidal symptoms (i.e. involuntary bodily movements), and studies have suggested that these agents have beneficial effects on a broader range of schizophrenia symptoms. In addition, many studies have shown that the atypical agents can decrease the length of hospital stays, increase compliance, and improve quality of life, compared with older typical agents. As a result, these “newer” atypical agents have gradually gained acceptance as first-line treatment options in consensus guidelines for schizophrenia.

Our physician consultants view the efficacy of the atypical agents as similar, despite disparate results from various company-sponsored clinical studies, and the relative side effect profiles have largely provided a means for differentiation. Reported side effects with these atypical agents include weight gain, metabolic syndrome, and several other individual metabolic/cardiovascular issues. Our consultant indicates that most centers will generally start with one of the more tolerable atypical agents (e.g. risperidone or aripiprazole), and then will move to a stronger dopamine blocker if patients do not have an adequate response, followed by augmentation with other therapies if necessary. Our consultants indicated to us that clozapine remains commonly used in “treatment-resistant” schizophrenia patients due to its favorable efficacy profile in this patient population, though side effects remain an issue with this medication, and could provide justification to use a newer atypical agent (e.g. ITCI’s Caplyta), especially in patients with certain co-morbidities like obesity.

Initial Atypical Antipsychotic Medications Face Significant Generic Competition; Development Now Focused On Long-Term Injectable Formulations

As many of the initial approvals of the “newer” atypical antipsychotic agents occurred more than 20 years ago, many of these products now face significant generic competition. These generic formulations have largely eroded sales of branded products, though several pharmaceutical companies have now developed long-acting injectable (LAI) formulations, which has provided another source of differentiation for these franchises. Our consultants indicate to us that these alternative formulations (which require much less frequent administration relative to oral products) have gained adoption among patients who respond to oral medication but who remain non-compliant or lack the motivation to take a once-daily treatment. However, these injectable options do not work for everyone, and side effects remain an issue with many of these products.

Otsuka’s Abilify (aripiprazole) is indicated for long-term maintenance of schizophrenia in adults and adolescents, maintenance and adjunctive therapy for manic and mixed episodes associated with bipolar I disorder, adjunctive therapy of MDD, and irritability associated with autistic disorder. Aripiprazole is viewed as the drug of choice for long-term treatment of schizophrenia given a tolerable side-effect profile despite lower efficacy than alternatives. Generic equivalents of aripiprazole were launched in April 2015 and have since taken the major share of the franchise. Abilify Maintena, Otsuka’s LAI formulation of Abilify, was approved in February 2013, and was the first long-acting injectable on the market.

In Q3:15, Alkermes launched Aristada (aripiprazole lauroxil), a once-monthly depot injection formulation of an aripiprazole pro-drug for the treatment of schizophrenia. Aristada is available in four doses (441mg, 662mg, 882mg, and 1,064mg) and can be administered via either gluteal or deltoid injection. In the U.S., Alkermes has initially targeted schizophrenia patients on oral aripiprazole for conversion to Aristada, with a secondary target of patients who take other oral antipsychotics. In June 2018, Aristada Initio (aripiprazole lauroxil nanocrystal dispersion) was approved by the FDA for the initiation of Aristada, in combination with a single oral dose of aripiprazole (once tolerance of oral aripiprazole has been established). The formulation of Aristada Initio, compared with normal Aristada, allows for more rapid achievement of therapeutic drug levels and, therefore, a shorter period of required pre-injection oral supplementation (a factor in the dosing regimen that has partly been a hurdle to long-acting injectable adoption), and does not require 3 weeks of concomitant oral supplementation. Alkermes reported FY2025 sales of Aristada and Aristada Initio of \$370MM, and the company expects 2026 sales to fall in the range of \$365-385MM.

Risperdal (risperidone) has garnered widespread use given its solid effectiveness and broad label and is often used as first-line therapy. FDA approved the first generic risperidone in 2008, and several other generic approvals followed. Despite concern among some physicians, Risperdal’s propensity to raise prolactin does not appear to be an important issue in most patients.

JNJ/Alkermes’s Risperdal Consta is a two-week intra-muscular LAI formulation of Risperdal and was the first approved depot formulation of an atypical antipsychotic for the treatment of schizophrenia and bipolar disorder. Risperdal Consta was launched in the EU in 2002 and in the U.S. in 2003. Despite the need for greater compliance on atypical antipsychotics, adoption of Risperdal Consta in the U.S. (53% of WW sales) was slow for multiple reasons: (1) the U.S. psychiatry community lacks the treatment infrastructure to effectively administer depot injections every two weeks (i.e., outpatient clinics for schizophrenia patients); (2) schizophrenic patients resist intra-muscular injections; (3) reimbursement has been difficult; and (4) Risperdal Consta may be under-dosed for the more severe patient population. However, reimbursement has improved since launch, and worldwide sales of Risperdal Consta reached \$592MM in 2021, though generics launched beginning in Q4:25.

TEVA and Medincell have launched **Uzedy**, a subcutaneous LAI risperidone formulation delivered every 1-2 months. The therapy was first approved for schizophrenia in 2023 following positive data demonstrating a relapse-free rate of 93% after 24 weeks (vs. 72% for those on placebo), with a label expansion to bipolar I disorder in 2025. Uzedy sales in 2025 were \$191MM.

Invega (oral paliperidone) is an active metabolite of risperidone and another atypical antipsychotic compound. JNJ launched Invega in 2007 as part of its franchise extension strategy. Our consultants indicate Invega is unique due to its limited effect on orthostatic hypotension and excellent efficacy in delirium, where it is the drug of choice. However, this aspect of differentiation has not translated into greater prescriptions. Invega sales have deteriorated to negligible levels.

In 2009, JNJ launched Invega Sustenna (paliperidone palmitate), a LAI form of Invega, for acute treatment and maintenance of schizophrenia in adults. The rollout of Invega Sustenna has expanded the market for LAI atypicals, though some cannibalization of Risperdal Consta prescriptions has occurred. Additionally, Invega Sustenna offers multiple administration advantages over Risperdal Consta, which include: (1) a four-week long-acting injection vs. two-week injection for Risperdal Consta; (2) a more rapid onset of activity; (3) no oral therapy supplementation required on initiation (easier titration); (4) a smaller-gauge needle (23 gauge) than required for Risperdal Consta (20 gauge); and (5) room temperature storage. JNJ has developed multiple additional formulations for Invega that continue to improve upon the convenience of the therapy, including Invega Trinza (a once-quarterly formulation launched in 2015) and Invega Hafyera (a twice-yearly formulation approved in 2021). In 2024 JNJ reported WW sales of \$4.2B, which declined to \$3.8B for the combined product franchise in 2025.

Lilly's Zyprexa (olanzapine) is approved for the treatment of schizophrenia, acute and maintenance monotherapy of bipolar disorder, adjunctive therapy with lithium or valproate for bipolar disorder, and agitation associated with schizophrenia and bipolar I mania. While efficacious, Zyprexa leads to significant weight gain, with ~22% of patients gaining at least 7% of their baseline weight over a median exposure of 8 weeks. Despite its metabolic profile, several generic olanzapine equivalents launched beginning in 2011 and have taken a substantial market share from the branded product. In 2009, Eli Lilly launched Zypadhera, Zyprexa's four-week LAI formulation, in several European countries, and in the U.S. as Zyprexa Relprevv in 2010. Zyprexa Relprevv can be initiated without oral supplementation, an advantage over JNJ/Alkermes's Risperdal Consta. However, our physician consultants prefer the use of Risperdal Consta due to the substantial metabolic disturbances associated with Zyprexa and Zyprexa Relprevv.

Teva/Medincell's TEV-749 is a long-acting extended-release once-monthly formulation of olanzapine currently under review with the FDA based on positive data from the Ph. III SOLARIS study on the PANSS at week 8. The data also demonstrated sustained improvement in CGI-S and PSP scale scores. Importantly, no cases of PDSS were observed in the clinical development program, which would provide a likely differentiated safety profile vs. olanzapine if approved. While ~36% of patients treated with the therapy experienced weight gain, the side effect accounted for <1% of study discontinuations.

AstraZeneca's Seroquel XR (quetiapine, extended release) is considered the most widely used and versatile antipsychotic and has a favorable side-effect profile. Seroquel is indicated for the treatment of schizophrenia, acute manic episodes associated with bipolar disorder, maintenance of bipolar disorder and depressive episodes associated with bipolar disorder. In addition to these indications, Seroquel was approved for MDD in the EU in 2010. Prior to the 2012 launch of quetiapine generics, Seroquel IR (immediate release) held a 23% U.S. prescription share of the antipsychotic market which declined to 1% in May 2013.

The availability of Seroquel IR generics led to a 15% Y/Y loss in Seroquel XR sales at the time. Seroquel XR generics then made their market debut in November 2016, leading to further market share deterioration of the branded version. In 2019, AstraZeneca sold off the commercial rights for Seroquel and Seroquel XR in the U.S. and Canada to Cheplapharm due to the loss of patent protection.

HLS Therapeutics' Clozaril (clozapine) is indicated for both treatment-resistant schizophrenia and reduction of suicidal behavior in patients with schizophrenia or schizoaffective disorder. The use of Clozaril and its generic equivalents has been limited due to the possibility of a life-threatening side effect, agranulocytosis, which occurs in ~1% of patients. Because of this risk, patients treated with clozapine are required to have weekly blood tests for the first six months on drug. This substantially limits clozapine's utility, as schizophrenia patients are notoriously difficult to monitor. Pharmacokinetic data have shown lower peak plasma concentrations of generic clozapine vs. Clozaril, suggesting that generic clozapine may result in fewer side effects than the branded drug. Generics currently comprise most of total clozapine prescriptions written.

Pfizer's Geodon (ziprasidone) is indicated for schizophrenia, acute agitation associated with schizophrenia, bipolar mania, mixed and manic episodes of bipolar disorder and adjunctive maintenance treatment of bipolar disorder in adults. Geodon's prescription share has eroded significantly since its March 2012 patent expiration, and we foresee a negligible amount of sales of the drug in the U.S. into perpetuity.

Sunovion's Latuda (lurasidone) was approved in 2010 for the treatment of patients with schizophrenia. Latuda was approved with a black box warning label for increased mortality in elderly patients and increased suicidal thoughts and behaviors. It has been suggested that the efficacy of lurasidone in schizophrenia is mediated through a combination of D2 and 5HT2A receptor antagonism. Latuda's exclusivity period ended in 2019, and in January 2019, the FDA approved multiple generics for lurasidone HCl.

Later Approvals Have Provided Additional Treatment Options, Though Differentiation Is Unclear

Given the unfavorable weight gain and metabolic side effects associated with many of the initial atypical antipsychotic medications, as well as the generally non-compliant nature of schizophrenia patients, treatment adherence remains a major issue in the schizophrenia space. As a result, patients will sometimes cycle through multiple different therapies before they find something that works, and in recent years, several additional products have entered the schizophrenia market. However, as these newer products all generally rely upon some form of dopamine antagonism in order to ameliorate schizophrenia symptoms, our consultants indicate that they have not significantly differentiated themselves from prior therapies to date, despite somewhat improved safety profiles compared to older antipsychotics.

AbbVie's Vraylar (cariprazine) received FDA approval in 2015 as a once-daily treatment for schizophrenia and for acute manic or mixed episodes associated with bipolar I disorder. In May 2019, the FDA expanded the Vraylar label for use in the treatment of bipolar depression. Vraylar is an atypical antipsychotic and its efficacy is thought to be mediated through a combination of partial agonist activity at central dopamine D2 and serotonin 5HT1A receptors and antagonist activity at serotonin 5HT2A receptors. The FDA based approval of Vraylar on results from three 6-week placebo-controlled studies, in which treatment with Vraylar demonstrated a placebo-adjusted 8.8 improvement on the PANSS score at week 6 at the 6 mg/day maximally recommended dose.

Vraylar (which has patent protection through 2028) has benefited from favorable clinician/patient experience, which our consultants have confirmed. In addition, anecdotal feedback from clinicians (as described by company management) has

indicated that the product can cause a “brightening” for their patients with negative symptoms, which has led to increased excitement for the product. Geodon Richter announced positive top-line Phase IIIb data of Vraylar for the treatment of schizophrenia patients with persistent and predominant negative symptoms in January 2015, but the FDA issued a refusal-to-file letter to Allergan in 2017 after it deemed the application not sufficiently complete for a substantive review. Worldwide sales of Vraylar were \$3.6B in 2025. Allergan was acquired by ABBV in 2020.

JNJ/Intra-Cellular's Caplyta (lumateperone) received FDA approval in December 2019 for the treatment of schizophrenia in adults and Intra-Cellular launched Caplyta commercially in late Q1:20. Caplyta is a 5HT_{2A} receptor antagonist that has both pre-synaptic and post-synaptic antagonist activity at the D₂ receptors and acts as a glutamatergic modulator and a serotonin reuptake inhibitor. The FDA based approval of Caplyta on results from two placebo-controlled trials. In the first study (a Ph. II study), treatment with Caplyta resulted in a statistically significant placebo-adjusted change of 5.8 points on the PANSS total score at week 4 at the recommended 42 mg/day dose.

In the second study (a Ph. III study), treatment with Caplyta at the 42 mg/day dose resulted in a statistically significant 4.2-point improvement in the PANSS total score at week 4. Intra-Cellular also studied the efficacy of Caplyta in a second Ph. III study in schizophrenia patients, but Caplyta did not produce a statistically significant effect relative to placebo in this study, which Intra-Cellular attributed at the time to an abnormally high placebo response.

In September 2017, Intra-Cellular announced topline data from 302 patients in Part A of the long-term safety study in schizophrenia in which patients switched from SOC treatment to a 42 mg dose of Caplyta for 6 weeks, then returned to SOC therapy. Data from 6 weeks of treatment strongly supported prior placebo-controlled data which suggested that Caplyta may have an improved metabolic, lipid and endocrine safety/tolerability profile vs SOC. Highly statistically significant improvements were seen in body weight, BMI, and waist circumference. Highly statistically significant improvements in lipid levels were also seen in LDL, total cholesterol, and triglycerides. Prolactin levels were also significantly reduced from baseline. Given this metabolic profile, KOLs continued to see a role for lumateperone in the treatment of schizophrenia, likely as maintenance therapy in certain patients, given its metabolic profile, despite the fact that Caplyta failed to meet the primary endpoint in the second Phase III trial. We note that the therapy is also approved for bipolar depression and as an adjunctive treatment for MDD. ITCI was acquired by JNJ in April 2025, and JNJ recorded \$700MM in sales last year.

Alkermes's Lybalvi (ALKS 3831) received FDA approval in June 2021 for the treatment of schizophrenia as well as treatment of bipolar I disorder (both as a monotherapy for maintenance treatment and as a monotherapy or adjunct to lithium or valproate for acute treatment of manic or mixed episodes). Lybalvi is a fixed dose oral combination of olanzapine and ALKS 33 (samidorphan, a proprietary opioid antagonist). The FDA based the approval of Lybalvi on one 4-week, randomized, controlled study in schizophrenia patients, as well as previously conducted controlled studies of oral olanzapine. In the aforementioned study with Lybalvi, treatment with Lybalvi at doses of 10 mg olanzapine/10 mg samidorphan or 20 mg olanzapine/10 mg samidorphan resulted in a placebo-adjusted improvement of 6.4 points on the total PANSS score at week 4. Alkermes also evaluated Lybalvi's weight gain profile head-to-head against olanzapine in a second, 24-week study. Treatment with Lybalvi in this study resulted in a statistically significant difference in the mean percent change from baseline body weight compared to olanzapine (4.2% vs 6.6% for olanzapine), as well as a lower percentage of patients who gained 10% or more of the baseline body weight (17.8% vs 29.8% for olanzapine).

Physicians generally like the Lybalvi data and appear especially interested in its more favorable metabolic profile. We note that Alkermes has looked to position the drug as particularly effective in patients with the dual diagnosis of comorbid substance

abuse, as ALKS 33 also potentially has non-addictive properties. Lybalvi sales were \$347MM in 2025, with guidance of \$380-400MM for 2026.

BMS/Karuna's Cobenfy Has Been Approved For The Treatment Of Schizophrenia

Karuna Therapeutics (acquired by BMS in March 2024) successfully developed Cobenfy (KarXT) for patients with schizophrenia and the agent was approved in September 2024. KarXT combines xanomeline, a muscarinic receptor agonist that preferentially stimulates M1 and M4 muscarinic receptors, and trospium, a muscarinic receptor antagonist. Trospium does not measurably cross the blood-brain barrier, which allows for confinement of the effects of xanomeline in peripheral tissues and preferentially stimulates the muscarinic receptors in the CNS.

In November 2019, Karuna announced the Phase II trial investigating KarXT for acute psychosis in patients with schizophrenia met the primary endpoint with an 11.6-point reduction in PANSS total score at week 5 vs. placebo ($p < 0.001$). Karuna evaluated KarXT as a monotherapy in its EMERGENT program, which consists of the completed positive registrational EMERGENT-1, EMERGENT-2 and EMERGENT-3 studies, and two ongoing Phase III long-term safety studies (EMERGENT-4 and EMERGENT-5).

Similar to EMERGENT-2 (the Ph. II study), the Ph. III EMERGENT-3 trial was a five-week in-patient trial evaluating the efficacy and safety of KarXT compared to placebo in $n=256$ adults with schizophrenia. Patients were randomized 1:1 to receive a flexible dose of KarXT ($n=125$) or placebo ($n=131$) BID for 5 weeks. In March 2023, Karuna announced that KarXT met its primary endpoint, demonstrating an 8.4-point placebo-adjusted reduction in the PANSS total score vs placebo (-20.6 KarXT vs -12.2 placebo, $p < 0.0001$) at week 5 (effect size: 0.60). Similar to the EMERGENT-2 data, treatment with KarXT in the study also resulted in statistically significant reductions in the total PANSS score compared to placebo at weeks 2, 3, and 4. Also similar to the EMERGENT-2 results, treatment with KarXT resulted in a 3.5-point placebo-adjusted reduction in the PANSS positive subscale (-7.1 KarXT vs -3.6 placebo, $p < 0.0001$) at week 5.

However, unlike the prior dataset, treatment with KarXT only demonstrated numerically higher reductions in the PANSS negative subscale at week 5 (-2.7 KarXT vs -1.8 placebo; $p=0.12$) as well as on the PANSS negative Marder factor subscale (-3.5 KarXT vs. -2.7 placebo; $p=0.20$), though it did result in statistically significant improvements on both measures at week 4 (due in part to a reduced placebo response at that time).

In terms of safety, the overall discontinuation rate was 33% (37% KarXT vs 29% placebo). TEAE rates for KarXT was 70% vs 50% for placebo. Discontinuation rates related to TEAEs were similar between treatment arms (6% KarXT vs 5% placebo), consistent with prior studies. One serious TEAE reported in the KarXT arm was related to gastroesophageal reflux disease and deemed not to be related to the study drug. The most common TEAEs were mild or moderate in severity, including nausea, dyspepsia, vomiting, constipation, headache, hypertension, diarrhea, and insomnia. Importantly, there were no discontinuations due to hypertension. Mean blood pressure measures were similar between KarXT and placebo, with no syncopal events reported. An increase in heart rate was associated with KarXT treatment but decreased in magnitude by the end of the study, consistent with prior trials. Karuna noted that vital signs are taken at 2-hours post-dose across all EMERGENT studies, which corresponds to the C_{max} of KarXT, where maximal effect on blood pressure is expected to be seen. Rates of weight gain, somnolence, and extrapyramidal symptoms were similar between KarXT and placebo.

In April 2025, BMS announced that a Ph. III study of Cobenfy as an adjunctive therapy to atypical antipsychotic treatment failed to meet the primary endpoint of change from baseline to week 6 in the PANSS total score. On the primary endpoint, the combination resulted in a reduction in the PANSS total score of 14.3 points vs. 12.2 points for the monotherapy arm ($p=0.11$). Interestingly, the therapy did show larger

benefit in patients receiving a non-risperidone atypical antipsychotic. Our KOLs have indicated that they would still use Cobenfy in combination with a background antipsychotic given the current standard of care; however, we will have to see how ease of reimbursement plays out. Cobenfy achieved 2025 sales of \$155MM and our pharma colleagues model \$3.1B in 2030.

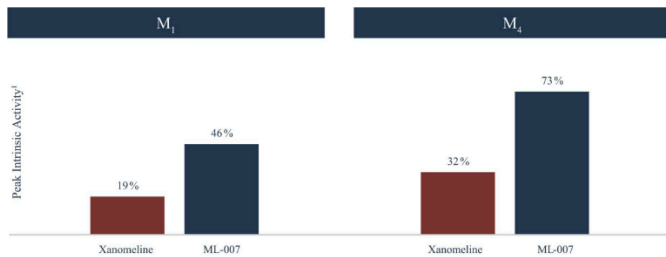
Maplight Is Advancing A Hopefully Improved Muscarinic For Schizophrenia

Despite the abundance of generic frontline treatment options and newer antipsychotic agents, our KOL consultants indicate that roughly 25-30% of schizophrenia patients do not have an adequate response to currently available products. Our consultants note that there remains a significant unmet need for new products with differentiated mechanisms of action that address a broad range of schizophrenia symptoms of the disease, which also avoid the typical tolerability issues associated with direct dopamine antagonism.

Maplight Therapeutics is developing ML-007C-MA, a fixed dose combination of a M1/M4 muscarinic agonist (ML-007), which is co-formulated with a peripherally acting anticholinergic (fesoterodine). Muscarinic receptors are a family of G protein-coupled receptors that are activated by acetylcholine, where they function in various neurophysiological pathways including those implicated in cognition, behavior, sensory, and motor function. Muscarinic receptors are found presynaptically at nerve endings to prevent the release of neurotransmitters; they are present in autonomic ganglia and there are five primary subtypes (M1-5). Due to the distribution of various muscarinic receptors across the brain and body, targeting this pathway could have implications across a broad range of neurological indications. In a simplistic way, an agonist targeting the muscarinic pathway imitates acetylcholine and its function as a neuromodulator. M1 receptors are found mostly in the stomach, salivary glands, and cerebral brain; M2 receptors are present mostly in the heart tissue and smooth muscle; M3 receptors are also present in the salivary glands, stomach, and smooth muscle; and M4 receptors are found in the substantia nigra and hippocampal regions of the brain, as are M5 receptors.

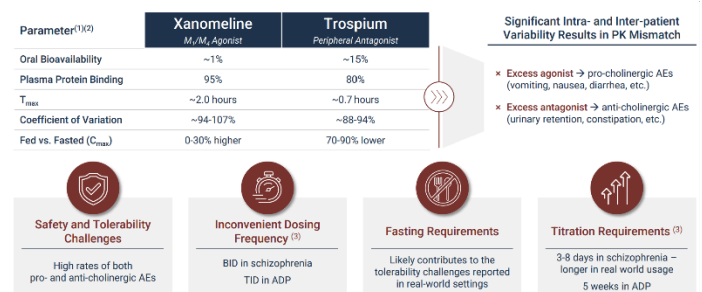
The company indicates that ML-007 is specifically designed to activate both M1 and M4 muscarinic receptors, to drive efficacy, as it is thought hitting both of these subtypes will be important to drive the overall neuropsychiatric benefit for patients, while of course avoiding receptors in non-implicated peripheral tissues. Specifically, the company notes targeting both receptors' functions to counterbalance dopamine activity through direct and indirect pathways to reduce delusions and hallucinations, while activation of M1 receptors specifically functions to enhance the formation of coherent and linear short- and long-term memories which can cause neurostability. Maplight indicates peak intrinsic activity of ML-007 is over 2x higher than xanomeline based on the GTP γ S assay, and believes that targeting M1 is important for learning and memory benefits not possible with M4-specific approaches. In addition, MPLT highlights that Cobenfy has fasting requirements (taken one hour or more before a meal, and two hours after a meal) and titration. Through improved targeting, MPLT is confident that it can improve upon Cobenfy's profile.

ML-007 Has Greater Relative Intrinsic Activity Vs. Xanomeline



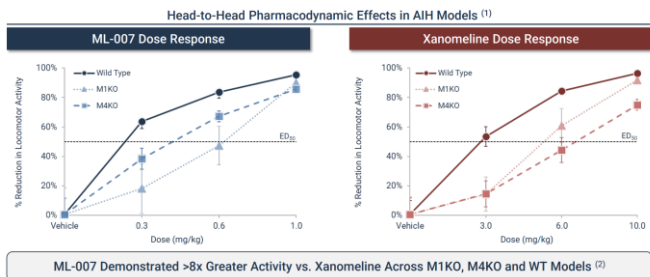
Source: Maplight Therapeutics

Maplight Sees Room For Improvement On Cobenfy



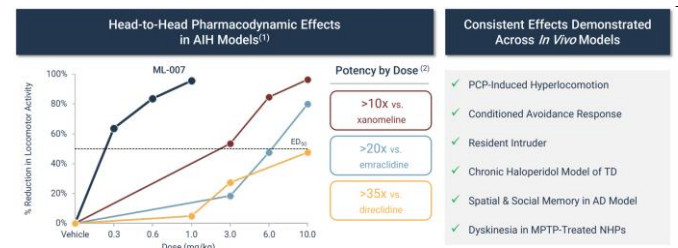
Maplight conducted pre-clinical testing in animal models to determine that the target efficacious CSF concentration range was 14-27 ng/mL. Benchmarking to xanomeline, across pre-clinical *in vivo* models, the company indicates ML-007 has 10-fold greater potency by dose. The company also demonstrated that activity at both M1 and M4 receptors vs. either individual receptor alone resulted in improved pharmacodynamic effects in KO mouse models of amphetamine-induced hyperlocomotion. As ML-007 has stronger activity at M1, the therapy demonstrated improvements in spatial and social memory in an Alzheimer's mouse model, whereas xanomeline did not.

ML-007 And Xanomeline In Mouse KO AIH Models



Source: Maplight Therapeutics

ML-007 AIH Model Potency Vs. Other Muscarinic Agents



ML-007 Improves Spatial/Social Memory In AD Mouse



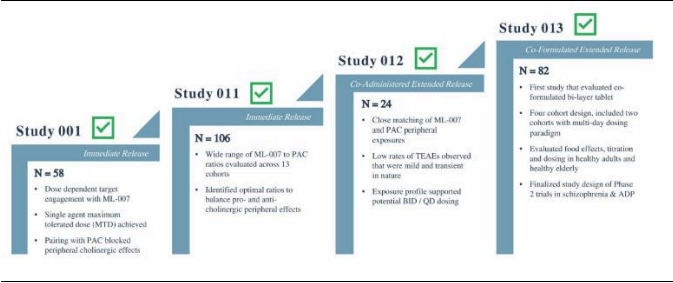
Source: Maplight Therapeutics

Past work with scopolamine, a non-selective muscarinic receptor antagonist, has induced psychotic symptoms. On the other side, prior experiments with more nonspecific muscarinic receptor agonist compounds have suggested that this class has the potential to address symptoms associated with schizophrenia. In addition, xanomeline, a muscarinic acetylcholine receptor agonist, demonstrated improvements in both cognition and psychotic symptoms, though resulted in AEs of nausea and vomiting. As cells express muscarinic receptors in several locations in the body, repeated systemic administration of agonists to these receptors has led to gastrointestinal side effects that have limited clinical development progress. Previous work has suggested that activation of the M2/M3 receptors play a large role in the occurrence of these gastrointestinal side effects due to the expression of these receptors in the smooth muscle, while the M1/M4 receptors have relatively higher expression in the CNS (with M4 receptor subtypes differentially expressed in the striatum). Recent research with selective M1 receptor modulators has suggested that compounds that target this receptor can still result in gastrointestinal toxicity despite limited activity at other receptor subtypes.

Initial Ph. I Clinical Work Helped To Set Dosing Profile And Demonstrate Safety

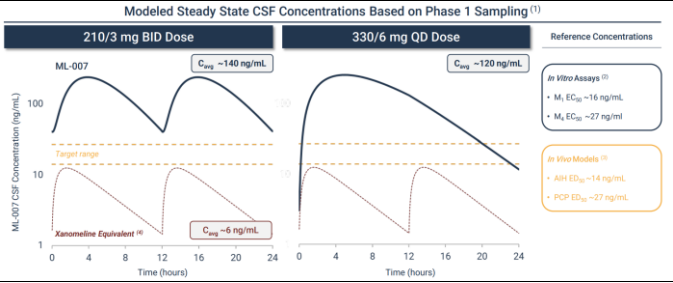
ML-007 was studied in four Ph. I trials including 270 HVs and >1.5K doses of drug. These trials include: **(1) Study 001**, a single agent ML-007 MTD trial that demonstrated the ability to block peripheral cholinergic effects with a PAC, **(2) Study 011**, which was designed to evaluate the peripheral side effect profile for the combination of 007 and the PAC. Across 13 cohorts and 22 dosing paradigms it was determined the target plasma ration for ML-007:PAC was 100:1 to 600:1 as levels less than this resulted in anticholinergic effects, and levels higher than this resulted in procholinergic effects. **(3) Study 012** examined extended-release profiles of ML-007 and the PAC, and **(4) Study 013** was the first trial to use the same formulation in the Ph. II (bi-layer, co-formulation tablet ML-007C-MA. The study evaluated doses up to 210/3mg BID and 330/6 QD in SAD and MAD cohorts of healthy adult/elderly participants. The trial demonstrated low PK variability at steady state and the ability to maintain plasma ratios of the component therapeutic parts to deliver a solid tolerability profile. The TEAEs that emerged in this trial are detailed below, and the company noted that no episodes of vomiting or constipation were reported and there were no episodes of anticholinergic TEAEs.

ML-007C-MA Ph. I Program

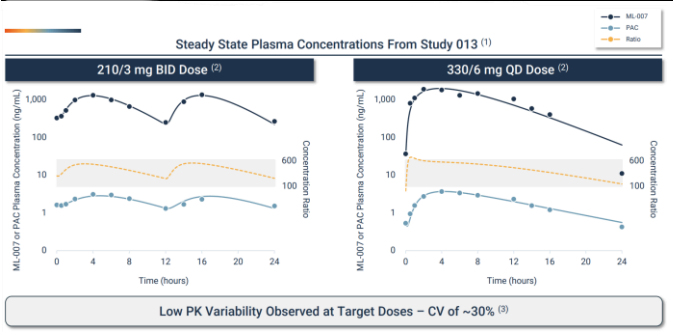


Source: Maplight Therapeutics

CSF Exposure Exceeds Target Range In Ph. I

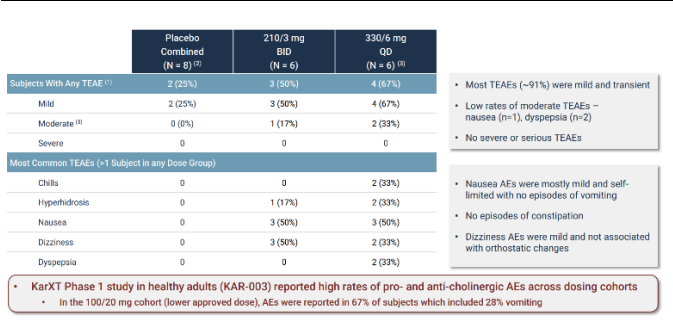


Matching Of Component Exposures Should Limit AEs



Source: Maplight Therapeutics

Ph. I (Study 013) TEAEs At Ph. II Target Dose



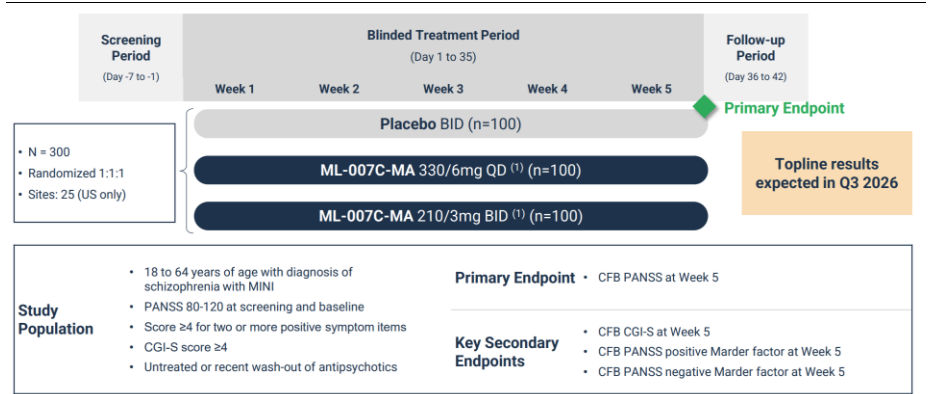
ML-007C-MA Is In A Ph. II Study, Expected To Yield Data In Q3:26

Based on the Ph. I dose exploration work, Maplight advanced ML-007C-MA into a Ph. II study for schizophrenia with data expected in Q3:26. The study will enroll n=300 hospitalized adults experiencing acute exacerbation of psychotic symptoms and underlying schizophrenia across 25 U.S. sites. Patients will be randomized to either placebo BID or ML-007C-MA 330/6mg QD or ML-007C-MA 210/3mg BID. The primary endpoint is the change from baseline vs. placebo on the PANSS at week 5. Patients on antipsychotic medications will need to discontinue the treatments during screening. Patients will receive a single titration dose of 105/3mg before moving to the maintenance doses above.

Management indicates that the study is 90% powered for an effect size of 0.5, which the company believes is conservative relative to what was reported by KRTX. The team notes that the KRTX study was likely functionally unblinded to some extent given the level of nausea and vomiting seen with the active drug; however, the initial KRTX study did not see the level of anticipation bias muscarinic trials will now, so that will need to be considered in the placebo arm.

The team notes that the Ph. II study size was designed to be registrational in nature, thus the findings should be compared to more traditional pivotal studies in the space vs. exploratory Ph. I/II outcomes, and we agree that is reasonable. Depending on the outcomes from the trial, MPLT indicates it could potentially move both dosing paradigms forward. In the QD arm, the therapy will see a higher C_{max}, and thus while MPLT continues to anticipate a superior tolerability profile even from this arm vs. Cobenfy, the BID arm may have even better tolerability, and thus patients can tailor their approach depending on outcomes. When examining tolerability, MPLT notes that it will be important to look at both frequency and overall severity of AEs.

Ph. II ZEPHYR Study Of ML-007C-MA In Schizophrenia



Source: Maplight Therapeutics

ML-055 Follows ML-007 In the Pipeline

In conjunction with the Q4:25 earnings report, MPLT announced an earlier pipeline candidate (ML-055), a next-generation M1/M4-muscarinic agonist for the treatment of neuropsychiatric indications. The company indicates that potential candidates have demonstrated increased potency pre-clinically, and a formal pre-clinical candidate to advance into IND-enabling studies will be selected in 2026. MPLT notes that this therapy is also more lipophilic vs. ML-007, and thus could be developed as a QD oral therapy, but also as a long-acting injectable formulation. As ML-007 is more soluble, a LAI formulation is not as feasible.

Additional Clinical Programs Are Advancing For The Treatment Of Schizophrenia

Despite the number of approved agents in the schizophrenia space today, several different product candidates are advancing in development for the indication. While several of these programs plan to target both positive and negative symptoms of schizophrenia, a number of companies have focused development specifically on the negative symptoms of the disease (which our consultants have suggested represents a larger opportunity due to the higher unmet need). We review some of the most advanced candidates in development for schizophrenia in the table below, with greater detail on muscarinic programs.

AbbVie/Cerevel's Emraclidine Comes Up Short In Ph. II Schizophrenia Program

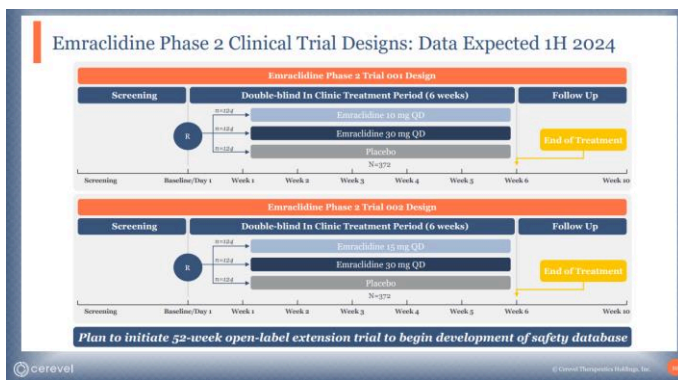
Cerevel's (acquired by AbbVie in August 2024) CVL-231 (emraclidine) is an M4-selective PAM being developed as a potential treatment for schizophrenia. Emraclidine selectively targets activation of the M4 receptor in the brain to reduce dopaminergic activity without direct dopamine receptor antagonist activity. By selectively targeting the M4 receptor, emraclidine has the potential to avoid the serious gastrointestinal side effects that have commonly been associated with, and have hindered clinical development of, non-selective muscarinic agents. Following encouraging Phase Ib data (~11.9 point pbo-adjusted reduction in the PANSS at week 6), Cerevel advanced emraclidine to two Phase II studies (EMPOWER-1 and EMPOWER-2) in June 2022.

The two Phase II program multi-dose, double-blind, placebo-controlled trials enrolled n=379 and 373 schizophrenia patients, respectively, with acute exacerbation of psychotic symptoms with baseline PANSS scores from 85-120 and CGI-S ≥4. Trial 001 randomized patients 1:1:1 to receive either once-daily 10 mg emraclidine, 30 mg emraclidine, or placebo, while Trial 002 randomized patients 1:1:1 to receive once-

daily 15 mg emraclidine, 30 mg emraclidine, or placebo. Each trial had a 6-week double-blind in clinic treatment period, followed by a 4-week follow up period. The primary endpoint in each study was change in PANSS total score after 6 weeks compared to baseline scores.

ABBV announced in November 2024 that both trials failed to meet the primary endpoint. The details on the primary endpoint are in the figure below. Based on these results ABBV is investigating emraclidine in a Ph. II adjunctive study for schizophrenia. MPLT indicates that it believes the prior Ph. I experience for emraclidine was perhaps not sufficient to design an effective pivotal program, and hitting only M4 leaves some efficacy on the table, and the therapy could be dose limited, leaving opportunity for ML-007 to provide a stronger efficacy profile.

Schematic Design Of Ph. II Trials For CVL-231 In Schizophrenia



Source: Cerevel Therapeutics

Change From Baseline in PANSS Score At Week 6 In Ph. II Studies

	EMPOWER-1			EMPOWER-2		
	Placebo (N = 127)	Emraclidine 10mg QD (N = 125)	Emraclidine 30mg QD (N = 127)	Placebo (N = 128)	Emraclidine 15mg QD (N = 122)	Emraclidine 30mg QD (N = 123)
Baseline (SD)	98.3 (8.16)	97.6 (7.65)	97.9 (7.89)	97.4 (8.22)	98.0 (8.49)	97.2 (7.75)
LS Mean (95% CI)	-13.5 (-17.0, -10.0)	-14.7 (-18.1, -11.2)	-16.5 (-20.0, -13.1)	-16.1 (-19.4, -12.8)	-18.5 (-22.0, -15.0)	-14.2 (-17.6, -10.8)

Source: AbbVie

Neurocrine Is Advancing Its Own Muscarinic Assets

Neurocrine initiated a placebo-controlled Phase II study evaluating a M4 agonist, direclidine (formerly NBI-1117568 or HTL-0016878, from Sosei Heptares), for the treatment of schizophrenia in October 2022. '568 is a 4th generation oral M4 agonist that avoids the cooperativity with ACh, improving efficacy and minimizing side effects. This molecule has been explored by Sosei in collaboration with Allergan in a Phase I study in healthy volunteers demonstrating a tolerable safety profile.

Data from the Ph. II study were released in August 2024. The once-daily 20 mg dose met the primary endpoint (change in PANSS total score from placebo) with a 7.5 point improvement in PANSS score compared to placebo ($p=0.011$) with a -18.2 point change in PANSS in 20 mg QD vs. -10.8 points in the placebo and effect size of 0.61. The trial did not show a dose response with higher doses (40 mg QD, 60 mg QD and 30 mg BID) not meeting either the primary or secondary endpoints at week 6. At the 20 mg QD dose, direclidine demonstrated a significant difference from placebo starting at week 3 with the response deepening through week 5. The greatest difference in PANSS score was at week 5 with a 10-point improvement from placebo with an effect size of 0.72.

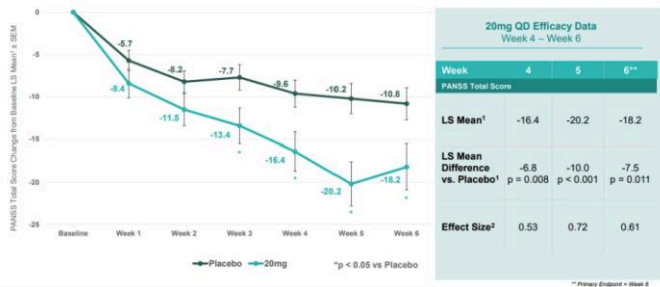
Neurocrine also evaluated clinical global impression scale severity (CGI-S), and Mardar Factor positive and negative symptom change (a model of PANSS) in the Ph II trial. The 20 mg QD dose hit statistical significance on all three endpoints demonstrating a CGI-S score improvement of 0.7 points ($p<0.0001$) a Mardar factor - positive score improvement of 3 points ($p=0.004$) and Mardar factor - negative score improvement of 1.9 points ($p=0.028$) vs. placebo.

Changes In PANSS Score Through Week 6 Across Doses

Week 6	Placebo N=68	20mg QD N=35	40mg QD N=38	60mg QD N=34	30mg BID N=26
PANSS Total Score					
LS Mean Change from Baseline*	-10.8	-18.2	-12.6	-13.7	-15.8
LS Mean Difference vs. Placebo, p-value*		-7.5 p = 0.011	-1.9 p = 0.282	-2.9 p = 0.189	-5.0 p = 0.090
Effect Size**		0.61	0.27	0.39	0.23

Source: Neurocrine

Impact Of Direclidine's 20mg Dose On PANSS Through Week 6



Source: Neurocrine

Impact on Secondary Endpoints

Week 6	CGI-S		Marder Factor — Positive		Marder Factor — Negative	
	Placebo N=68	20mg QD N=35	Placebo N=68	20mg QD N=35	Placebo N=68	20mg QD N=35
LS Mean Change from Baseline*	-0.5	-1.2	-2.8	-5.8	-1.2	-3.1
LS Mean Difference vs. Placebo*		-0.7 p < 0.001		-3.0 p = 0.004		-1.9 p = 0.028

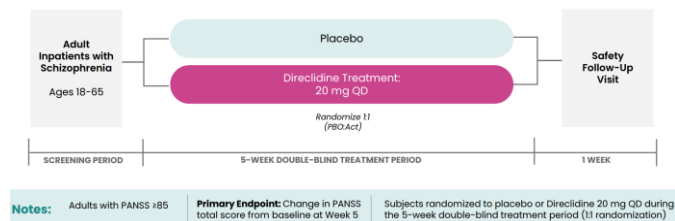
In terms of safety, direclidine was well tolerated with minimal GI events and no weight gain relative to placebo. The most common AEs were somnolence (11% in all treated vs. 3% in placebo), dizziness (9% vs. 1%), headache (9% vs. 20%), nausea (6% vs. 3%), and constipation (5% vs. 3%) and none appeared to be dose dependent. Discontinuation rates across treated and placebo arms were comparable (5% vs. 4.3%). Management noted that there was a low incidence of CV events including hypertension or static hypotension and they were not dose related. Additionally, there were few extrapyramidal symptom AEs reported.

While the results support advancement of direclidine, the data fell somewhat short of investor expectations. Investors were most focused on direclidine's placebo-adjusted PANSS score and were hoping for a placebo-adjusted change of at least 10 points to be competitive with the Ph. Ib and II data from BMY/Karuna's KarXT (11.6 point placebo-adjusted change in PANSS in Ph. II) and ABBV/Cerevel's emraclidine (12.7 point placebo-adjusted change in PANSS in Ph. Ib). Moreover, direclidine produced an inverse dose response, with higher doses performing more poorly than the 20mg dose. This has caused investors to further question the robustness of direclidine's data.

Nonetheless, management has provided reasons to be optimistic that direclidine's efficacy is competitive. NBIX notes that the change from baseline for 20mg direclidine (-18.2 points) is similar to the Ph. Ib or II data from KarXT (-17.4 points) and emraclidine (-19.5 points) and that the difference compared to those earlier trials was a larger placebo effect in the NBIX study. In fact direclidine's data is similar to KarXT's Ph. III data (placebo-adjusted PANSS changes of -8.4 and -9.6) where the placebo effects (-11.6 and -12.2) were similar to those observed in the NBIX Ph. II. Additionally, 20mg direclidine's effect size (0.61) is similar to that of KARXT's Ph. IIIs (0.60-0.61).

Neurocrine is conducting two short-term schizophrenia studies that will enroll an estimated n=280 patients randomized 1:1 to either 20mg of direclidine or placebo. The primary endpoint will be the change in the PANSS. Topline data readouts for the program are expected in 2027-2028.

Direclidine Ph. III Short-Term Trial Design



Source: Neurocrine

Direclidine Pivotal Program In Schizophrenia

Study Description	Number of Subjects Per Study	Topline Data Expected
Short-Term (Acute) Study #1	284	2027
Short-Term (Acute) Study #2	284	2028
Randomized Withdrawal	560	2029
Long-Term Safety (Open Label)	800	2031

Neurocrine has several other muscarinic agents in development across indications with '570 in Ph. II development and '569/567 in Ph. I.

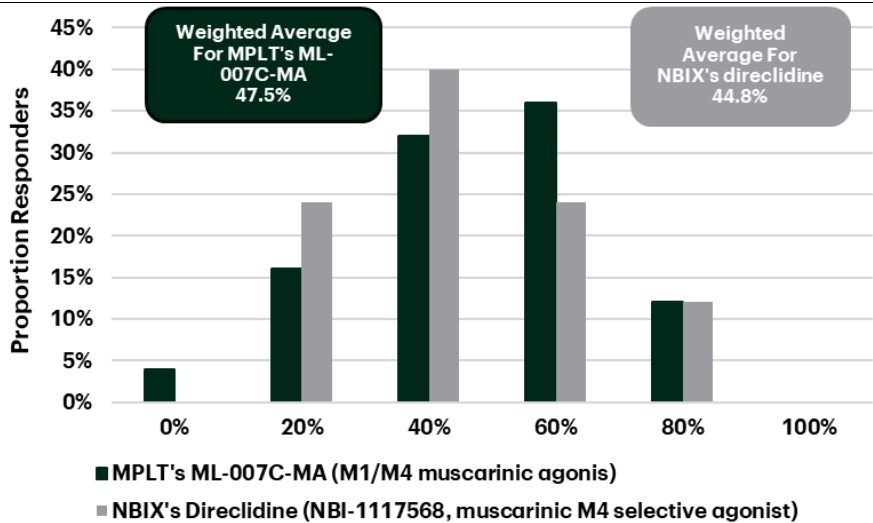
Neurocrine's Pipeline Of Muscarinic Agents

	Direclidine	NBI-'570	NBI-'569	NBI-'567
Mechanism of Action	M4 Agonist	Dual M1 / M4 Agonist	Dual M1 / M4 Agonist	M1 Preferring Agonist
Half Life	11 – 16 Hours	18 – 22 Hours	35 – 50 Hours	17 – 19 Hours
Dosing and Administration	Once Daily	Once Daily, LAI Potential	Once Daily	Once Daily
Indication	Schizophrenia & Bipolar Mania	Schizophrenia	Alzheimer's Psychosis	Alzheimer's Cognition & LBD

Source: Neurocrine

We asked n=25 physician respondents their predicted probability of success for the ongoing studies of MPLT's ML-007C-MA and NBIX's direclidine on the primary endpoint of change from baseline vs. placebo on the PANSS score at five weeks, and while responses varied, a weighted average for the Maplight compound was 47.5% vs. 44.8% for the Neurocrine therapy.

Physicians Predict Slightly Higher POS For Ongoing Study Of MPLT's ML-007C-MA Compared To NBIX's Direclidine



Source: TD Cowen, March 2026 physician survey, n=25

Select additional therapies in development for the treatment of schizophrenia are detailed in the chart below.

Other Notable Programs In Early Commercial Phase Or Clinical Development For The Treatment Of Schizophrenia

Product	Company	Approach/MOA	Delivery	Status	Comments
Bysanti	Vanda Pharm.	5-HT2A, dopamine D2, and alpha-1 adrenergic receptor antagonist	Oral adjunctive therapy	Approved; Product availability in Q3:26	Demonstrated bioequivalence to iloperidone; also approved for bipolar disorder
Roluperidone	Minerva Neurosciences	5-HT2A receptor antagonist	Oral	Ph. III for negative symptoms of schizophrenia	Data expected in H2:27
Balipodect	Axsome	PDE10A	Oral	Ph. III-ready	Acquired from TAK; Ph. III-enabling studies to begin in 2026.
Briloroxazine	Reviva Pharm.	Dopamine-serotonin modulator	Oral	Ph. III complete, FDA recommended another Ph. III study	Ph. III trial initiation planned for mid-2026
NMRA-898	Neumora	M4 PAM	Oral	MAD study in schizophrenia ongoing	Data expected in H2:26

Source: TD Cowen, Company Disclosures

Managing Placebo Response Is Key For Success In Neuropsych Studies

Due to the subjective nature of many patient assessment metrics in the neuropsychiatry space, coupled with the often-complex disease presentation of patients with mental health concerns, increased “anticipation” bias, and potential benefits from more regimented therapy intervention, many patients in clinical trials in the space can show improvement on some metrics even after treatment with placebo

alone. Thus, effective clinical trial design and mitigation of placebo response is often key for success in these studies, including for schizophrenia. A table of historical outcomes including placebo rates is detailed in the chart below, but MPLT management acknowledges the importance of this, and has implemented several measures to help limit the response in placebo patients.

These measures include limiting the number of sites to the U.S., with conservative enrollment targets with a focus on enrolling patients with moderate-to-severe symptoms, making sure the clinical trial sites are well-vetted with regular staff training and making sure the duplicate or "professional" patients are not included in the trial, and independent central review and blinded data analysis. We find it encouraging that management acknowledges the importance of managing for placebo response, and we are hopeful these efforts will be rewarded in the Ph. II study.

Historical Active And Placebo PANSS Outcomes In The Clinical Trial Setting

Therapy	Trial	NCT number	Endpoint	Time	Placebo	Active	Hit on Primary?
Lybalvi	Ph. III ENLIGHTEN-1	NCT02634346	Total PANSS	Week 4	-17.5	-23.9	Yes
Caplyta	Ph. II study	NCT01499563	Total PANSS	Day 28	-7.4	-13.2	Yes
Caplyta	Ph. III study	NCT02282761	Total PANSS	Day 28	-10.3	-14.5	Yes
Caplyta	Ph. III study	NCT02469155	Total PANSS	Week 6	-15.1	-14.6	No
KarXT	Ph. II EMERGENT-1	NCT03697252	Total PANSS	Week 5	-5.9	-17.4	Yes
KarXT	Ph. III EMERGENT-2	NCT04659161	Total PANSS	Week 5	-11.6	-21.2	Yes
KarXT	Ph. III EMERGENT-3	NCT04738123	Total PANSS	Week 5	-12.2	-20.6	Yes
Ulotaront (75 mg)	Ph. III DIAMOND-1	NCT04072354	Total PANSS	Week 6	-19.3	-19.6	No
Ulotaront (75 mg)	Ph. III DIAMOND-2	NCT04092686	Total PANSS	Week 6	-14.3	-16.4	No
Brilaroxazine (50 mg)	Ph. III RECOVER	NCT05184335	Total PANSS	Week 4	-13.8	-23.9	Yes
GWP42003	Ph. II study	NCT02006628	Total PANSS	Week 6	-8.8	-11.2	No
Emraclidine (30mg)	Ph. II EMPOWER-1	NCT05227690	Total PANSS	Week 6	-13.5	-16.5	No
Emraclidine (30mg)	Ph. II EMPOWER-1	NCT05227703	Total PANSS	Week 6	-16.1	-14.2	No
Emraclidine (30 mg QD)	Ph. Ib study	NCT04136873	Total PANSS	Week 6	-6.8	-19.5	Yes
NBI-1117568 (20 mg)	Ph. II study	NCT05545111	Total PANSS	Week 6	-10.8	-18.2	Yes
Pimavanserin	Ph. III ENHANCE	NCT02970292	Total PANSS	Week 6	-13.4	-15.3	No

Source: TD Cowen, Company Disclosures

KOLs See Potential For The Muscarinic Pathway And Room For Differentiation From Cobenfy

We discussed the role that newer muscarinic-based agents would play in the schizophrenia treatment paradigm with our KOL consultants. Both of our KOLs highlighted the unmet need for additional agents for schizophrenia that do not directly target dopamine receptors as part of the mechanism of action, which could provide patients who do not respond to currently available therapies with another treatment option. While payors will most likely relegate use of these novel agents to patients who have failed previous treatment options, one of our KOLs noted that he would "love" to start treatment of schizophrenia patients with a non-D2 antagonist due to the superior tolerability profile that these newer agents could provide.

With respect to Cobenfy, our KOLs have suggested that uptake has been slower than they would have liked primarily due to P&T committee restrictions which restricted Cobenfy access to patients with intolerable extrapyramidal symptoms. The physicians noted that this is partially likely due to the development path Karuna pursued which enrolled patients more refractory and post therapies that block D2, so it has been known largely as a drug to be used in the refractory setting. It was our KOLs' wish that these therapies be used more in early-course patients and that development programs can reflect that.

In addition, one KOL clarified that while the trospium is intended to block the GI distress, nausea, and vomiting caused by xanomeline, the trospium is not well absorbed, especially if one has taken the pill after breakfast. While one should take the therapy on an empty stomach, the patient population is not necessarily the most adherent to medication schedules, so the real-world GI consequences have been a

bit more challenging. If a physician puts a patient on Cobenfy, and they have a challenging GI side effect profile, that will color that physician's opinion of the therapy and impact prescribing. While the KOL expects this can be managed over time, a therapy that does not have this same profile, or more challenging administration schedule, could see improved adoption.

Our physicians have indicated that Cobenfy has "opened the door", and that typically what happens is the first therapy enters the market, and then those behind it can improve on the tolerability or dosing profile. There remains "much opportunity" to improve upon Cobenfy's profile, in particular regarding the food effect-our physician KOL said it's a bit counterintuitive in that in general, patients who take their medications on a full stomach have fewer side effects, but in the case of Cobenfy, the opposite is true.

While the physicians note that the eventual patient data will be the determinant of ML-007C-MA's competitive profile, a therapy that could be dosed more easily either through QD or without consideration to food would be an advancement.

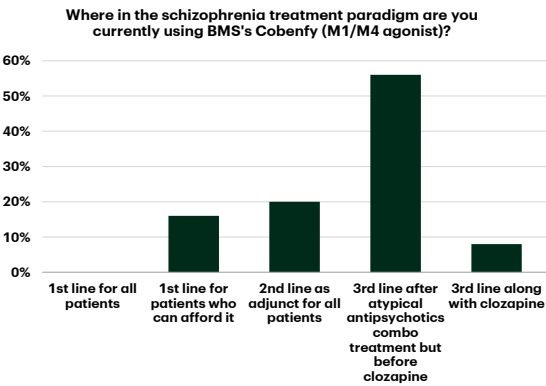
We Expect ML-007C-MA Could Be A Blockbuster, Sharing The Market With Cobenfy

Maplight is hopeful that success in the ongoing Ph. II study could count as one of two pivotal trials necessary for approval. If the data are supportive, the company will launch two additional Ph. III studies for schizophrenia (initiating one extra study has become commonplace in the neuropsych space given the potential variability in outcomes. Should the first two trials be successful, the third can likely be modified to gain additional information in discrete subgroups). Companies in the space generally assume that the FDA will require 300 patients followed for six months and 100 patients followed for 52 weeks ahead of a regulatory approval.

We estimate that there are ~2MM schizophrenia patients in the U.S. Of these patients, our consultants indicate that roughly 20-30% of patients do not receive treatment for their condition. Due to the abundance of low-cost generic options for the treatment of schizophrenia, we expect that payors will initially relegate use of ML-007C-MA to patients who do not have a satisfactory response to one or more currently available therapies and need additional augmentation/monotherapy options. Of the treated patients, we estimate that 50% only have a partial response to current schizophrenia medications and could use additional augmentation therapy. Due to the potentially favorable tolerability profile of ML-007C-MA compared to other approved or developmental compounds for the treatment of schizophrenia and the compelling muscarinic mechanism, we believe that ML-007C-MA could potentially see use in earlier lines of therapy as an add-on or adjunctive treatment for these patients, though we note that Maplight has not yet committed to conducting a specific study in this population. In fact, our KOLs note that a study such as this may not be prudent given reimbursement of combination therapies is likely to occur without clinical trial validation. We model an initial approval in 2029 for ML-007C-MA.

In our March 2026 physicians survey (n=25), 56% of our respondents indicated that BMS's Cobenfy is currently used in the 3L after atypical antipsychotic combination treatment but before clozapine. 16% indicated that they used it 1L for patients that had access and could afford it, and 20% indicated that it was used 2L as an adjunct for all patients. Only 8% indicated that it was used 3L along with clozapine.

Majority Of Physicians Currently Use BMS's Cobenfy 3L After Atypical Antipsychotic Combinations



Source: TD Cowen March 2026 Health Care Conference (n=25 physicians)

We expect ML-007C-MA will split the market with other approved muscarinic agents, but that a compelling differentiated profile will emerge through the clinical program. We model peak penetration of 5% in the adjunctive setting and 13% in the monotherapy later line setting, with a launch price of \$19K prior to gross to net discounting. This results in a peak U.S. revenue estimate of \$1.3B in 2040. We assume an ex-U.S. partnership resulting in ~\$1.5B in peak annual revenues to MPLT. This opportunity results in a value of \$35/share on an NPV basis.

ML-007C-MA Schizophrenia NPV (\$MM)

Financial Year End	12/31
Valuation Date	4/6/2026
Discount Rate	12%

\$MM	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E
U.S. 007 Sales	0	0	0	36	165	358	614	860	1050	1084	1125	1167	1211	1256	1302
Growth (%)		0%	0%	0%	362%	117%	72%	40%	22%	3%	4%	4%	4%	4%	4%
Royalty Payments From Partners	0	0	0	0	10	32	79	134	151	154	156	159	162	164	167
Total Revenues	0	0	0	36	175	390	693	994	1201	1238	1281	1326	1372	1420	1469
Growth (%)		0%	0%	0%	389%	123%	78%	43%	21%	3%	4%	3%	3%	3%	3%
COGS	0	0	0	4	17	36	61	86	105	108	113	117	121	126	130
COGS as a % of sales	0%	0%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
R&D	153	225	250	165	125	75	50	20	10	0	0	0	0	0	0
R&D as a % of Revenues	0%	0%	0%	462%	72%	19%	7%	2%	1%	0%	0%	0%	0%	0%	0%
SG&A	47	60	65	125	250	295	305	310	315	320	325	325	325	325	325
SG&A as a % of Revenues	0%	0%	0%	350%	143%	76%	44%	31%	26%	26%	25%	25%	24%	23%	22%
Operating Income	-200	-285	-315	-258	-217	-16	276	578	771	809	844	884	926	969	1014
Tax	0	0	0	0	0	0	55	116	154	162	169	177	185	194	203
Tax rate	0%	0%	0%	0%	0%	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%
Approx Free Cash Flow	(200)	(285)	(315)	(258)	(217)	(16)	221	462	617	648	675	707	741	776	811
Years	-2.27	-1.27	-0.27	0.73	1.73	2.73	3.73	4.73	5.73	6.73	7.73	8.73	9.73	10.73	11.73
Discount Factor	1.29	1.15	1.03	0.92	0.83	0.74	0.66	0.59	0.53	0.47	0.42	0.38	0.34	0.30	0.27
NPV of Cash flows	(257)	(328)	(325)	(238)	(179)	(12)	146	273	326	306	286	268	251	235	220

Terminal Value Calculation

Final year FCF	811
Perpetual Growth Rate	-100%
Terminal Value	0.0
Discount Factor	0.27
Present Value of Terminal Value	0.0
Present Value of Cash Flows	1,039.8
Enterprise Value	1,039.8
Market Value	1,492.9
Fully Diluted Shares Outstanding	42.4
Value per Fully Diluted Share	\$35

Source: TD Cowen

ML-007C-MA Schizophrenia Market Model (\$MM)

U.S. Market	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E
Estimated # patients with schizophrenia ('000)	2000	2010	2020	2030	2040	2051	2061	2071	2081	2092	2102	2113
% of diagnosed & treated patients	76%	77%	78%	79%	80%	81%	82%	83%	84%	85%	86%	87%
# diagnosed and treated schizophrenia patients	1520	1548	1576	1604	1632	1661	1690	1719	1748	1778	1808	1838
% of patients who could benefit from additional augmentation then	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%
# patients who could benefit from additional augmentation therapy	760	774	788	802	816	830	845	859	874	889	904	919
% penetration of 007	0%	1%	3%	4%	5%	5%	5%	5%	5%	5%	5%	5%
# patients on 007 as adjunctive therapy	2	10	20	28	37	42	42	43	44	44	45	46
% compliance	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%
Price per patient	\$19,000	\$19,000	\$19,000	\$19,380	\$19,768	\$20,163	\$20,566	\$20,978	\$21,397	\$21,825	\$22,262	\$22,707
Price per patient after GTN	\$11,400	\$11,400	\$13,300	\$13,566	\$13,837	\$14,114	\$14,396	\$14,684	\$14,978	\$15,278	\$15,583	\$15,895
007 adjunctive sales in schizophrenia (\$MM)	\$16	\$86	\$196	\$286	\$381	\$440	\$456	\$473	\$491	\$509	\$528	\$548
# diagnosed and treated schizophrenia patients	1520	1548	1576	1604	1632	1661	1690	1719	1748	1778	1808	1838
% of patients that don't respond to current treatments	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
# patients with insufficient response to current treatments	456	464	473	481	490	498	507	516	525	533	542	551
# patients who received 007 in earlier line	0	2	10	20	28	37	42	42	43	44	44	45
# patients who have not received 007 and need additional therapy	456	462	463	461	462	462	465	473	482	490	498	506
% penetration of 007	1%	2%	4%	7%	10%	13%	13%	13%	13%	13%	13%	13%
# patients on 007 as adjunctive therapy	2	9	16	32	46	58	58	59	60	61	62	63
% compliance	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%
Price per patient	\$19,000	\$19,000	\$19,000	\$19,380	\$19,768	\$20,163	\$20,566	\$20,978	\$21,397	\$21,825	\$22,262	\$22,707
Price per patient after GTN	\$11,400	\$11,400	\$13,300	\$13,566	\$13,837	\$14,114	\$14,396	\$14,684	\$14,978	\$15,278	\$15,583	\$15,895
007 monotherapy sales in schizophrenia (\$MM)	\$19	\$79	\$162	\$329	\$479	\$611	\$628	\$652	\$676	\$701	\$727	\$754
Total U.S. 007 sales (\$MM)	\$36	\$165	\$358	\$614	\$860	\$1,050	\$1,084	\$1,125	\$1,167	\$1,211	\$1,256	\$1,302
E.U. Market	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E
Estimated # patients with schizophrenia ('000)	2350	2362	2374	2385	2397	2409	2421	2433	2446	2458	2470	2483
% of diagnosed & treated patients	76%	77%	78%	79%	80%	81%	82%	83%	84%	85%	86%	87%
# diagnosed and treated schizophrenia patients	1786	1819	1851	1884	1918	1952	1986	2020	2054	2089	2124	2160
% of patients who could benefit from additional augmentation then	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%
# patients who could benefit from additional augmentation therapy	893	909	926	942	959	976	993	1010	1027	1045	1062	1080
% penetration of 007	0%	0%	1%	3%	5%	5%	5%	5%	5%	5%	5%	5%
# patients on 007	0	3	9	28	48	49	50	50	51	52	53	54
% compliance	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%
Price per patient	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300
007 adjunctive sales in schizophrenia (\$MM)	\$0	\$32	\$92	\$282	\$478	\$487	\$495	\$504	\$512	\$521	\$530	\$539
# diagnosed and treated schizophrenia patients	1786	1819	1851	1884	1918	1952	1986	2020	2054	2089	2124	2160
% of patients that don't respond to current treatments	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
# patients with insufficient response to current treatments	536	546	555	565	575	585	596	606	616	627	637	648
# patients who received 007 in earlier line	0	0	3	9	28	48	49	50	50	51	52	53
# patients who have not received 007 and need additional therapy	536	546	552	556	547	538	547	556	566	575	585	595
% penetration of 007	0%	0%	1%	2%	4%	5%	5%	5%	5%	5%	5%	5%
# patients on 007 as adjunctive therapy	0	2	7	11	19	27	27	28	28	29	29	30
% compliance	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%
Price per patient	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300	\$13,300
007 monotherapy sales in schizophrenia (\$MM)	\$0	\$16	\$69	\$111	\$191	\$268	\$273	\$277	\$282	\$287	\$292	\$297
Total EU 007 sales (\$MM)	\$0	\$48	\$161	\$393	\$669	\$755	\$768	\$781	\$795	\$808	\$822	\$835
Royalty to MPLT	\$0	\$10	\$32	\$79	\$134	\$151	\$154	\$156	\$159	\$162	\$164	\$167
W.W. 007 sales to MPLT (\$MM)	\$36	\$175	\$390	\$693	\$994	\$1,201	\$1,238	\$1,281	\$1,326	\$1,372	\$1,420	\$1,469

Source: TD Cowen

ML-007C-MA Is Also In Development For Alzheimer's Disease Psychosis

Due to the beneficial effects of targeting the muscarinic pathway on positive symptoms in schizophrenia patients, several companies are also investigating therapeutics for the treatment of psychosis in Alzheimer's disease patients. Literature sources suggest that roughly 30-40% of the ~7MM diagnosed Alzheimer's disease patients suffer from psychosis, which also correlates with increased caregiver burden, decreased quality of life, and earlier progression to nursing home care, severe dementia, and death. Often, physicians prescribe antipsychotics off-label for the treatment of psychosis in dementia patients. However, meta-analyses of randomized, controlled trials of anti-psychotics have suggested that these agents have limited efficacy in this patient population. Additionally, data have suggested that current antipsychotics can be associated with a significant decline in cognitive function in dementia patients with psychosis.

Given the lack of effective treatment options for this condition, Maplight investigated ML-007C-MA in its Ph. I program in healthy elderly patients (as part of Study 013), and a Ph. II study is now ongoing. In the Ph. I, both a two- and seven-day titration period was investigated with maintenance doses of 165/3 mg BID, 210/3 mg BID, and 330/6 mg QD used. It was noted that the titration dose of 105/1.5mg BID and

maintenance doses of 165/3 mg BID and 210/3 mg BID were well tolerated, but the QD dose was not well tolerated. At the go-forward Ph. II dose, the most common TEAEs are outlined in the table below, though notably there was no urinary retention, and only one episode each of vomiting and constipation.

Tolerability Profile In Healthy Elderly Patients

2-7 Day Titration + 7 Days at Target Dose

- Generally favorable safety and tolerability with BID dosing
- Highest dose given QD was not well tolerated – potential to explore lower doses in future
- Mostly mild and transient TEAEs
- Low rates of anticholinergic TEAEs
- No clinically meaningful changes in mean blood pressure or ECG parameters
- No liver-related findings

	Placebo Combined (N=7) ⁽¹⁾	165/3 mg BID (with 2d titration) (N=6)	210/3 mg BID (with 2-7d titration) (N=11) ⁽¹⁾	330/5 mg QD (with 7d titration) (N=6) ⁽¹⁾
Subjects with Any TEAE ⁽¹⁾	5 (71%)	4 (67%)	8 (73%)	6 (100%)
Cholinergic Classification – Subjects (%)				
Procholinergic	1 (14%)	2 (33%)	6 (55%)	6 (100%)
Anticholinergic	1 (14%)	1 (17%)	4 (36%)	0
Other	3 (43%)	3 (50%)	8 (73%)	6 (100%)
% of TEAEs by Severity				
Mild	100%	100%	93%	72%
Moderate ⁽¹⁾	0%	0%	7%	28%
Severe	0%	0%	0%	0%

Phase 2 Target Dose for ADP

Source: Maplight Therapeutics

Go-Forward ADP Dose Tolerability Profile

	Placebo Combined N = 7 ⁽²⁾	210/3 mg BID (with 2-7d titration) N = 11 ⁽²⁾
Subjects with Any TEAE ⁽¹⁾	5 (71%)	8 (73%)
Mild	5 (71%)	8 (73%)
Moderate	0	2 (18%)
Severe	0	0
Most Common TEAEs (≥2 Subjects)		
Hyperhidrosis	0	3 (27%)
Nausea	0	3 (27%)
Dizziness	1 (14%)	4 (36%)
Headache	0	3 (27%)
Tremor	0	3 (27%)

- Most TEAEs (~93%) were mild and transient
- Low rates of moderate TEAEs
- No serious or severe TEAEs

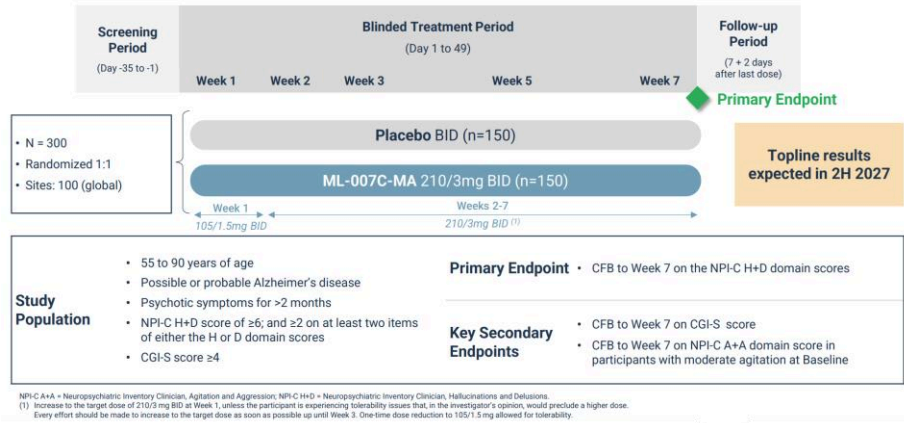
- Limited episodes of vomiting (n=1) or constipation (n=1)
- No episodes of urinary retention

KarXT Phase 1 study in healthy elderly volunteers (KAR-030) reported 47% of participants had ≥1 moderate AEs related to study drug (vs. 8% pbo), with higher rates of cholinergic AEs than previously reported in non-elderly

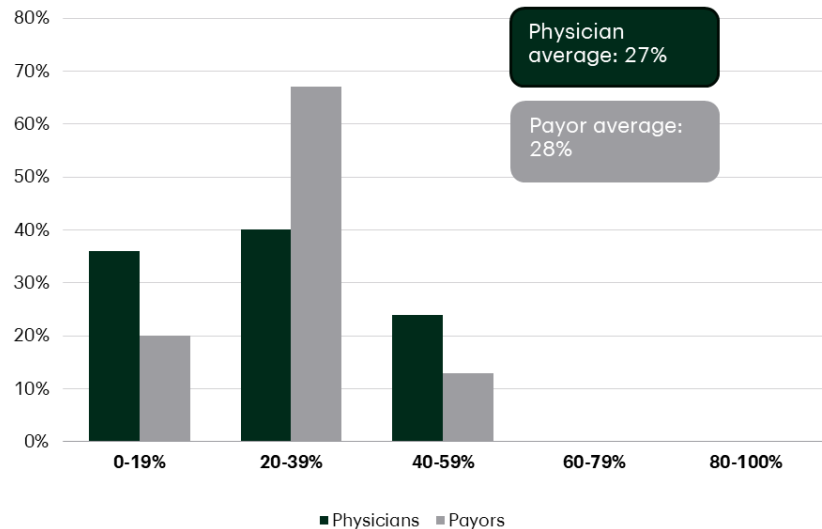
Ph. II VISTA Study Ongoing With Data Expected In H2:27

Maplight initiated a Ph. II trial of ML-007C-MA in patients with Alzheimer's psychosis that will randomize n=300 patients 1:1 across 100 global sites. Patients will receive a titration dose of 105/1.5mg BID for a week and then a target maintenance dose of 210/3mg BID for weeks 2-7. The primary endpoint is the change from baseline to week 7 on the NPI-C hallucination and delusion domains. Secondary endpoints include the CGI-S and the NPI-C A+A score in patients with baseline agitation. Data from the study are expected in H2:27.

Ph. II VISTA Study Of ML-007C-MA In Alzheimer's Psychosis



What Proportion Of Your Alzheimer's Disease Patients Under Care Exhibit Signs Of Psychosis And Would Benefit From Chronic Therapy?



Source: TD Cowen, 2025 CNS physician survey (n=25 physician, n=10 payor)

Cobenfy Is Also In Development For Alzheimer's Psychosis

BMS is also investigating Cobenfy in a development program for Alzheimer's psychosis. Based on results from a Ph. I HV study in elderly adults, the Ph. III ADEPT trials were initiated. The Ph. III ADEPT-1 study will enroll approximately n=380 patients with moderate or severe delusions/ hallucinations (NPI-C: D ≥ 2 or NPI-C: H ≥ 2) and CGI-S ≥ 4 . The study will consist of two phases: a single-blind, 12-week treatment period with Cobenfy for all patients, followed by a double-blind, 26-week randomized withdrawal period in which patients will either receive placebo (and therefore be withdrawn from treatment) or continue with Cobenfy treatment. In order to qualify as a "responder" in the first phase of the study, patients must exhibit a $\geq 40\%$ reduction in NPI-C: H+D score compared to baseline as well as a CGI-C score of 1 or 2 at both weeks 10 and 12, which should hopefully isolate clear responders to Cobenfy as opposed to pseudo-responders who only transiently meet the criteria for response. Similarly, in order to meet the criteria for relapse in the second phase of the study, patients must exhibit a 40% increase in the NPI-C: H+D score as well as a CGI score of 6 or 7, which BMS had hoped would allow it to more accurately measure a clinically meaningful effect of Cobenfy in this patient population. Other relapse criteria include treatment with other anti-psychotic agents, withdrawal from the study due to lack of efficacy, or hospitalization/worsening psychosis. The primary endpoint of the study will be time from randomization to relapse in patients treated with Cobenfy and placebo in the double-blinded randomized withdrawal phase of the study, while secondary endpoints will include time from randomization to discontinuation in either arm of the study. Patients who complete the ADEPT-1 trial will have the option to enroll in the long-term safety extension study (ADEPT-3). Data from the study are expected this year.

A second study (ADEPT-2) will evaluate the safety and efficacy of Cobenfy as an acute treatment for ADP compared to placebo in a more traditional double-blind placebo-controlled fashion. This study will randomize approximately 400 patients with a combined delusions and hallucinations score of ≥ 6 with either moderate-to-severe delusions or hallucinations (NPI-C: D ≥ 2 or NPI-C: H ≥ 2) and CGI-S ≥ 4 to receive either Cobenfy or placebo for a 14-week treatment period. Similar to the ADEPT-1 study, initial treatment with Cobenfy will proceed based on a flexible dosing and

titration schedule of Cobenfy TID. Patients who receive placebo will receive matching placebo TID. Also, like the ADEPT-1 trial, patients who complete the ADEPT-2 trial will have the option to enroll into the long-term safety extension study. The primary endpoint in the study will be change from baseline in the NPI-C: H+D score at the end of the treatment period, while secondary endpoints will include the change from baseline at the end of the treatment period in CMAI score, CGI-S, NPI-C core score, NPI-C/NPI: caregiver distress scale, NPI-C: agitation domain score, and number of NPI-C: H+D responders. BMS has also initiated the Ph. III ADEPT-4 trial for Cobenfy in ADP, which has a nearly identical design to the ADEPT-2 trial. Data from ADEPT-2 and -4 are also expected in 2026.

Notably in December 2025, BMS indicated that a blinded review of the ADEPT-2 study data identified irregularities in clinical trial execution at a small number of study sites. The company made the decision to exclude findings from those sites in the primary analysis prior to database lock, and in consultation with FDA decided to conduct an interim data analysis for efficacy and safety through an independent party. BMS remains blinded and decided to enroll more patients to account for those excluded in the analysis.

Cobenfy Clinical Program In Alzheimer's Psychosis

Indication	Psychosis in Alzheimer's Disease (ADP)			
Phase/Study	Phase III – ADEPT-1	Phase III – ADEPT-2	Phase III – ADEPT-4	Phase III – ADEPT-5
# of Patients	N = 380	N = 400	N = 406	N = 292
Design	- Cobenfy 20 mg/2 mg TID, 30 mg/3 mg TID, 40 mg/4 mg TID, 50 mg/5 mg TID, 66.7/6.67 mg TID* - Placebo	- Cobenfy 20 mg/2 mg TID, 30 mg/3 mg TID, 40 mg/4 mg TID, 50 mg/5 mg TID, 66.7/6.67 mg TID* - Placebo	- Cobenfy 20 mg/2 mg TID, 30 mg/3 mg TID, 40 mg/4 mg TID, 50 mg/5 mg TID, 66.7/6.67 mg TID* - Placebo	- Cobenfy BID* for Randomized Treatment and OLE periods: 56/6 mg, 112/12 mg, 168/18 mg, 224/24 mg - Placebo
Endpoints	- Primary: Time from randomization to relapse during the 26-week double blind randomized withdrawal period - Key secondary: Time from randomization to discontinuation for any reason during the 26-week Double-Blind Randomized Withdrawal treatment period	- Primary: Change from baseline in Neuropsychiatric Inventory–Clinician: Hallucinations and Delusions (NPI-C: H+D) score to end of Week 14 - Key secondary: Change from baseline in Clinical Global Impressions–Severity (CGI-S) scale up to Week 14	- Primary: Change from baseline in NPI-C: H+D score up to Week 14 - Key secondary: Change from baseline in CGI-S scale up to Week 14	- Primary: Change from baseline to end of Week 12 in NPI-C: H+D score - Key secondary: Change from baseline to end of Week 12 in CGI-S score
Status	- Recruiting - Projected data readout 2026	- Recruiting - Projected data readout 2026	- Recruiting - Projected data readout 2026	- Trial initiating - Projected data readout 2028
CT Identifier	NCT05511363	NCT06126224	NCT06585787	NCT06947941

Source: BMS

In addition, **ABBV's emraclidine** is in Ph. I development for the treatment of Alzheimer's psychosis, as is **NBIX's NBI-569**. Outside of the muscarinic class, **Acadia** has had a challenging path with the FDA regarding the approvability of pimavanserin for Alzheimer's psychosis, despite being FDA approved for Parkinson's psychosis. The company has decided to prioritize a next-generation 5-HT_{2A} inverse agonist candidate (**remlifanserin**, ACP-204) for Alzheimer's psychosis which is progressing in a Ph. II/III program with initial data expected in the August-October 2026 timeframe. The company anticipates that the next-generation therapy will allow for increased potency vs. pimavanserin without any QT prolongation risk, and an earlier onset.

The program includes a single Ph. II and two Ph. III trials with nearly identical design. The Ph. II trial will enroll n=318 patients randomized to either 30mg or 60mg remlifanserin or placebo. The primary endpoint in the study is the change from baseline to week 6 on the scale for the assessment of positive symptoms-hallucinations and delusions subscales (SAPS-H+D) total score. The Ph. II/III enrollment is expected to be seamless.

JNJ, through its acquisition of Intra-Cellular, is investigating **ITI-1284**, a deuterated form of lumateperone in a Ph. II study for Alzheimer's psychosis. The Ph. II study will enroll approximately n=370 patients with Alzheimer's disease and signs of psychosis (as measured by a score of 2 or greater on any item from Part A and/or Part B of the BEHAVE-AD scale) to receive either a 10 mg or 20 mg dose of ITI-1284 or placebo for a period of 6 weeks. The primary endpoint will be the change in the BEHAVE-AD psychosis subscale score from baseline at week 6, while the key secondary endpoint will be change in the CGI-S score from baseline to week 6. ITCI management had previously noted that it designed the trial as potentially registrational, and the posted trial record lists September 2027 as the estimated primary completion date.

We note that ADP represents a substantial potential market for ML-007C-MA. At this point the opportunity represents upside to our estimates as initial efficacy measures come into focus. As mentioned previously in this report, companies are investigating therapies targeting the muscarinic class in a diverse set of neuropsychiatric conditions. Maplight indicates that while it will look to establish efficacy and safety in schizophrenia and ADP initially, additional indications including cognition in Alzheimer's disease, Parkinson's disease psychosis, bipolar disorder, AD agitation, and autism spectrum disorder are areas of potential investigation in the future.

Representative additional trials of Cobenfy are detailed in the following chart.

Additional Clinical Studies Of Cobenfy Outside Of Schizophrenia And Alzheimer's Psychosis

Indication	Manic Episodes in Bipolar I Disease			Adjunctive Bipolar Mania		Alzheimer's Disease Agitation		Alzheimer's Disease Cognition (ADC)		Pediatric Autism Irritability								
Phase/Study	Phase III - BALSAM-1		Phase III - BALSAM-2		Phase III - BALSAM-4		Phase III - ADAGIO-1		Phase III - ADAGIO-2		Phase III - MINDSET 1		Phase III - MINDSET 2		Phase III - CN012-0044		Phase III - CN012-0045	
# of Patients	N = 274		N = 274		N = 424		N = 352		N = 352		N = 586		N = 586		N = 176		N = 176	
Design	• KarXT BID* • Placebo		• KarXT BID* • Placebo		• KarXT BID* + Background Treatment (LI, VPA, or Lamotrigine) • Placebo + Background Treatment (LI, VPA, or Lamotrigine)		• KarXT + KarX-EC • Placebo		• KarXT + KarX-EC • Placebo		• KarXT + KarX-EC • Placebo		• KarXT + KarX-EC • Placebo		• KarXT + KarX-EC • Placebo		• KarXT + KarX-EC • Placebo	
Endpoints	• Primary: Change from baseline in Young Mania Rating Scale (YMRS) at Week 3 • Key secondary: Change from baseline in Clinical Global Impressions-Bipolar (CGI-BP) at Week 3		• Primary: Change from baseline in YMRS at Week 3 • Key secondary: Change from baseline in CGI-BP at Week 3		• Primary: Change from baseline in YMRS total score at Week 5 • Key secondary: Change from baseline in Global Impression-Severity (GSI-S) score at Week 5		• Primary: Mean change from baseline on the Cohen-Mansfield Inventory-International Psychogeriatric Association (CMAI-IPA) at Week 14 • Key secondary: Mean change from baseline on CGI-S at Week 14		• Primary: Mean change from baseline on CMAI-IPA at Week 14 • Key secondary: Mean change from baseline on CGI-S at Week 14		Co-Primary: • Change from baseline in Alzheimer's Disease Assessment Scale-Cognitive Subscale 11 (ADAS-Cog11) at Week 24 • Clinician's Interview-Based Impression Plus Caregiver Input (CIBIC-PI) at Week 24 Key secondary: • Change from baseline in ADAS-ADL at Week 24		Co-Primary: • Change from baseline in ADAS-Cog11 at Week 24 • CIBIC-PI at Week 24 Key secondary: • Change from baseline in ADAS-ADL at Week 24		Primary: • Change from baseline in the Aberrant Behavior Checklist Irritability (ABC-I) Score at Week 8 Key secondary: • Change from baseline in CGI-S Score at Week 8		Primary: • Change from baseline in ABC-I Score at Week 8 Key secondary: • Change from baseline in CGI-S Score at Week 8	
Status	• Recruiting • Projected data readout 2027		• Recruiting • Projected data readout 2027		• Recruiting • Projected data readout 2028		• Recruiting • Projected data readout 2029		• Recruiting • Projected data readout 2028		• Recruiting • Projected data readout 2028		• Recruiting • Projected data readout 2028		• Trial initiating • Projected data readout 2029		• Trial initiating • Projected data readout 2029	
CT Identifier	NCT00951698		NCT00951711		NCT01746913		NCT01011732		NCT01011745		NCT00976216		NCT00976203		NCT01284745		NCT01285798	

Source: BMS

ML-004 Is In Ph. II Development For Autism Spectrum Disorder

Autism spectrum disorder (ASD) is a neurodevelopment disorder that is diagnosed by deficits in social communication in tandem with restricted interests and repetitive behaviors. In the U.S., ASD is estimated to affect about 3.2% of adolescents, although prevalence estimates have fluctuated due to changes in diagnostic criteria (Hodges et al, Transl. Pediatr. 2020; CDC ADDM). Maplight places these numbers at ~1.8MM children and >5MM adults in the U.S. In addition to differences in social communication and repetitive behaviors, emotional dysregulation and other behavioral problems are also common. Irritability is a broad term that encompasses reduced control over temper or mood resulting in verbal or behavioral outbursts and includes aggression and self-harm. The reported prevalence of irritability in individuals with ASD is variable, ranging from 19%-80%, but is associated with poorer quality of life across several domains including emotional and physical functioning as well as caregiver, social strain and success in behavioral intervention and general education. Extreme cases of irritability pose a significant threat to the individual (self-injury behavior) or others (aggression). Aggression towards others is reported in ~25-35% of ASD individuals and self-injury behavior may be as high as 42% of ASD individuals (Manter et al, BMC 2025).

There are two drugs approved in the U.S. for irritability associated with ASD, both part of the atypical antipsychotic class: Risperdal (risperidone), which was approved in 2006 and Abilify (aripiprazole), approved in 2009. While both drugs have proven to be effective at reducing irritability, the side-effect profile of atypical antipsychotics, including weight gain and extrapyramidal side effects, among others, can be challenging to manage in individuals with ASD. In addition, these agents are ineffective at treating the core social features of the disorder. Drug development in ASD has been challenging due to the varying symptomatology of patients, challenging diagnosis criteria, and lack of validated outcome measures.

ML-004 Looks To Address Social Communication Deficits And Irritability In ASD

ML-004 is 5-HT_{1B/1D} agonist, an IR/ER formulation of zolmitriptan (approved for the treatment of acute migraine). It has long been thought that serotonin dysregulation is implicated in the neurodevelopment of individuals with ASD, as reports as early as 1961 note abnormal 5-HT levels in the bloodstream of children with autism. Additional studies detailed in Yang et al. (Neurosci Bull, 2022) note that availability of the 5-HT transporter is lower in the brain of adults with ASD, and pre-clinical mouse experiments have displayed that alterations in serotonin signaling result in social impairments and communication deficits. MDMA has been shown to boost self-confidence and enhance sociability in humans. While the drug poses a meaningful risk of addiction, its mechanism has helped to substantiate the potential role of 5-HT and its circuitry in the pathogenesis of ASD.

ML-004 looks to drive similar impacts on sociability, without the addictive liability. Maplight conducted five *in vivo* animal studies including one focused on locomotion, and four evaluating zolmitriptan's impact on sociability and aggression in ASD model systems. In a mouse model, mice that are born following administration of valproic acid to the mother have a higher rate of social deficits. Treating these mice with 10 mg/kg zolmitriptan significantly reversed this social deficit. The same dose of zolmitriptan also significantly improved the social deficit phenotype of CNTNAP2 KO mice.

Maplight had to reformulate zolmitriptan in order to allow for chronic dosing at levels higher than those used for acute migraine treatment. A high-dose IR zolmitriptan oral formulation was studied in a Ph. I MAD PK study in n=40 HVs across 5 dosing cohorts (5mg, 10mg, 20mg, 30mg, 40mg) or placebo-dosed TID. Most AEs were mild with no SAEs and no clinically significant or dose-responsive changes in mean QTc. The main AEs were headache, dizziness, nausea, and hiccups which occurred at a higher frequency at the highest dose tested.

ML-004 (bi-layer IR/ER gastroretentive 24 mg oral formulation; 6mg IR/18mg ER) was studied in an open-label Ph. I study in n=12 HV adults with a dosing regimen of 20mg zolmitriptan IR on day 1, 24mg ML-004 under fasted conditions on day 2, and 24mg ML-004 under fed conditions on day 4. The single 24mg dose of ML-004 resulted in prolonged absorption and reduced C_{max} compared to IR zolmitriptan. The company notes that modeled data from this study and the pre-clinical experience suggest that ML-004 48mg and 72mg would result in the target plasma exposure for 12-15 hours, respectively. AEs in the study were less common with ML-004 dosing vs. IR zolmitriptan.

The Ph. II IRIS study of ML-004 is currently ongoing and enrolled n=160 adolescent (12-17 years, n=100) and adult (18-45 years, n=60) individuals with ASD. The placebo-controlled study uses a flexible dosing paradigm targeting maintenance doses of 48mg and 72mg. The primary endpoint is the change from baseline vs. placebo on the ABI (Autism Behavior Inventory) social communication domain score after 12 weeks of maintenance dosing.

Maplight notes that the Aberrant Behavior Checklist-Irritability (ABC-I) score is a key secondary endpoint for patients who have moderate or greater irritability at baseline. Topline results from the trial are expected in Q3:26. The company indicates that there are pre-defined subgroups in the study so that MPLT can interrogate any signals fully pending the final data, and that an eventual approval would likely have label differentiation vs. atypical antipsychotic agents, which feature a black box.

Success in ASD represents upside to our model, as we await additional details of ML-004's efficacy on the social components of ASD and safety profile, given the challenging development landscape historically.

KOLs Emphasize That New Therapies Would Be Readily Adopted, Though Development Has Been Challenging

We spoke with a leading KOL on the therapeutic treatment and development landscape for ASD. The physician noted that the need for therapies to treat behaviors associated with ASD is high. Diagnosis can be a process, as waitlists to see specialists can be up to a year, and evaluations can take a full day. Typically caregivers seek out a diagnosis due to signs of related conditions including ADHD, which can impact up to 50% of the ASD population. Symptoms of ASD do change with age, with social communication and daily living deficits in younger patients and increased signs of emotional dysregulation as the children get older.

Due to the side effect profiles of available treatments to treat agitation and aggression in ASD, therapeutic intervention tends to be second line following behavioral therapy. Many patients do end up on treatments for ADHD, but again it is a balance between the efficacy of stimulant-based options and the side effects. The KOL noted that all patients likely should at least trial a stimulant and see if it works for them; however, non-stimulant options should then be considered for those who find initial therapies intolerable.

A treatment for the social components of ASD would be useful for all patients, per our KOL, as he notes that over time, these are present in nearly everyone. The physician did note that the rating scales typically used in clinical studies are challenging as the disease presents in different ways patient to patient. That said, the social communication domain score of the ABL is reasonable. Any statistically significant change on this measure would be clinically meaningful for patients per our KOL given the lack of available treatments. An outcome that would be particularly compelling would be movement on several scales known to be correlated with disease severity including the Vineland, the CGI, and the visual analogue scale. While the KOL (who is familiar with assessment scales for ASD) noted that the FDA likes the Vineland, it is challenging to see meaningful improvement here given the multi-factorial nature of the assessments. For example, movement of the overall scale may depend on particularly pronounced movement on the social relationship domain.

In terms of the trial, the physician believes that the mechanism makes sense, as the serotonin pathway is definitely involved in the presentation of these behaviors. In addition, 12 weeks does seem like the appropriate timepoint for assessment. The KOL noted that studies longer than this period may suffer from attrition, while those that are shorter may not give the therapy sufficient time to work.

Other Product Candidates

In addition to the product candidates described above, Maplight has several other compounds in early clinical stages and preclinical development for various neuropsychiatric conditions:

ML-021 is a M4 selective antagonist currently in IND-enabling studies ahead of clinical investigation in Parkinson's disease. Maplight notes that the loss of striatal dopamine in PD causes an increase in striatal acetylcholine; this causes activation of the M4 receptors on neurons of the direct pathway, and contributes to motor deficits in the disease. The company notes that an M1 antagonist (which could be a target of a non-specific anticholinergic therapy), could cause cognitive impairment, psychosis, and GI constipation, thus the targeted M4 approach could limit off-target AEs. The company will finalize a candidate to advance into IND-enabling studies in 2027.

ML-009 is a GPR52 positive allosteric modulator (PAM) for investigation in hyperactivity/impulsivity disorders. GPR52 is a receptor expressed on indirect pathway neurons in the striatum. Activation of the receptor opposes the action of D2 receptors and was shown pre-clinically to reduce stimulant-induced hyperactivity, impulsivity, and aggression. Maplight has honed in on a pre-clinical candidate and expects to complete IND-enabling studies this year.

Intellectual Property

The composition of matter patents related to ML-007 expire in the 2031-2032 timeframe before any patent term extension. Additional pending patents related to the combination of ML-007 with the PAC for the treatment of schizophrenia and crystalline polymorphs expire between 2042 and 2045 before patent term extension.

For ML-004, pending patent applications related to the IR/ER formulation of the therapy and the associated use in ASD would expire in 2042 if granted prior to extensions. Pending patent applications associated with ML-009 and the GPR52 agonist portfolio would expire in 2044 before patent term extension, if granted.

Maplight Management Team With Solid Biotechnology Experience

The Maplight management team features a number of experienced veterans from the pharmaceutical/biotech space as well as individuals with specialized experience with neuropsychiatry/neurodegenerative diseases.

Chris Kroeger, M.D. is Maplight's CEO and was previously CEO of Cardioxyl Pharmaceuticals, which was acquired by BMS. Prior to this position, he was CEO of OvaScience, which merged with Millendo Therapeutics.

Erin Pennock Foff, M.D., Ph.D. is Maplight's current Chief Medical Officer. Prior to joining Maplight, she was the executive director of clinical development at Acadia and held academic positions focused on cognitive neurology.

James Lillie, Ph.D. is Maplight's current Chief Scientific Officer. Prior to joining Maplight, he was the CSO of OvaScience and was a VP of *in vitro* biology at Sanofi Genzyme.

Vishwas Setia is Maplight's current Chief Financial Officer. He was previously a Managing Director in the Bank of America investment banking group.

ESG Discussion

Maplight Therapeutics is a biotechnology company focused on the development of small molecule therapies for several neurological and neuropsychiatric conditions. Maplight's small molecule therapies have novel mechanisms of action that target the key neural circuitry in several CNS conditions and have the potential to provide additional therapeutic benefit to patients beyond currently available options. In addition, Maplight's therapies have lower specificity for protein receptor subtypes known to cause certain side effects in patients with these conditions, which should improve the tolerability profile of the treatment. Maplight Therapeutics' platform therefore has the potential to significantly improve the standard of care for several CNS indications with high unmet need. The company's lead programs include therapies for schizophrenia, Alzheimer's agitation, and autism spectrum disorder, and all have the potential to substantially reduce the burden of disease in these indications, which has clear benefits to society.

Maplight Therapeutics R&D Pipeline

Therapeutic Class/Product	Indication	P-C	I	II	III	FILING	MKT	Comments
ML-007C-MA	Schizophrenia			•				Data in Q3:26
ML-007C-MA	Alzheimer's Disease Psychosis			•				Data in H2:27
ML-004	Autism Spectrum Disorder			•				Data in Q3:26
ML-021	Parkinson's Disease	•						IND-enabling studies to begin in 2027
ML-055	Neuropsychiatric Indications	•						Candidate nomination in 2026
ML-009	Hyperactivity/Impulsivity	•						IND-enabling studies to complete in 2027
		3+	0	3	0	0	0	
Redwood City, CA		IR Contact: Vish Setia (224) 803-8453						

Source: TD Cowen



TD COWEN ESG SCORES

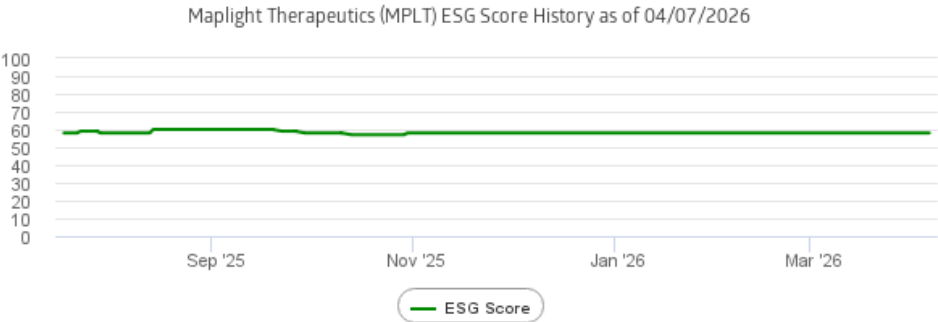
Maplight Therapeutics
NASDAQ: MPLT

ESG Score: 58/100

ESG Industry Percentile: 39th

Maplight Therapeutics(MPLT) ESG Material Category Rankings as of April 7, 2026

Top 3 Material ESG Categories	Dynamic Materiality TM	Score
Access and Affordability	43%	60
Business Ethics	20%	57
Customer Welfare	17%	62



ESG performance is scored on a 0 to 100 scale. A score of 50 represents a neutral impact. Scores above 50 indicate more positive performance, and scores below reflect more negative performance. A blank chart means the company has an ESG performance score of N/A. A score of N/A means not enough data is available on the company to generate a score.
Source: Truvalue Labs

ESG MATERIALITY

Establishing **materiality** is critical to evaluating a company's ESG performance. Factors most material in one sector (or to a particular company) may not be as important to another. In addition, the factors that are material – and the degree to which factors are material – can change over time.

Applying data to frameworks established by SASB (the Sustainability Accounting Standards Board) and by FactSet, we present in the chart above the three most material ESG factors that investors should focus on for the company that is the subject of this report; the Dynamic Materiality™ of each factor (i.e., what percentage of overall materiality the category represents for the subject company); and a Score for the subject company in each of these three categories (on a 0 to 100 basis, with 50 being average).

We also calculate an **overall ESG Score** for the subject company, which is presented above (in green) and on the cover of this report. A full explanation of how this ESG Score is derived is presented below.

HOW ARE TD COWEN'S ESG SCORES CALCULATED?



TD Cowen leverages technology from FactSet to generate our ESG scores. Truvalue uses artificial intelligence to capture the stakeholder view of how companies are performing on ESG metrics, using the Sustainability Accounting Standards Board (SASB) materiality framework (www.sasb.org). These data are leveraged to calculate a score for each company, which allows TD Cowen to have a **common framework** and uniform way to approach ESG discussions with our clients. TD Cowen ESG scores appear on Company and Company Quick Take notes and are updated daily.

HOW DOES THE PROCESS WORK?

The process begins with capturing unstructured data from more than 200,000 sources, in 38 languages. These data are culled from a wide range of sources with varied perspectives, including industry publications, news outlets, NGOs, trade unions, government sources, legal and regulatory filings, and academic publications.

Natural language processing is used to interpret semantic content from the original sources and generate analytics by applying criteria consistent with established sustainability and ESG frameworks. Performance is scored on a 0 to 100 scale. **A score of 50 represents a neutral impact.** Scores above 50 indicate more positive ESG performance, and scores below reflect more negative ESG performance. A score of NA means not enough data is available on the company to generate a score.

VALUATION METHODOLOGY AND RISKS

Valuation Methodology

Biotechnology:

In calculating our 12-month target price, we employ one or more valuation methodologies, which include a discounted earnings analysis, discounted cash flow analysis, net present value analysis and/or a comparable company analysis. These analyses may or may not require the use of objective measures such as price-to-earnings or price-to-sales multiples as well as subjective measures such as discount rates.

We make investment recommendations on certain early stage, pre-revenue companies based upon an assessment of their business model, technology, probability of market success, and the potential market opportunity, balanced by an assessment of applicable risks. Such companies may not be assigned a price target.

Investment Risks

Biotechnology:

There are multiple risks that are inherent with an investment in the biotechnology sector. Beyond systemic risk, there is also clinical, regulatory, and commercial risk. Additionally, biotechnology companies require significant amounts of capital in order to develop their clinical programs. The capital-raising environment is always changing and there is risk that necessary capital to complete development may not be readily available.

Risks To The Price Target

There are multiple risks that are inherent in an investment in the biotechnology sector. Beyond systemic risk, there is also clinical, regulatory, and commercial risk. Additionally, biotechnology companies require significant amounts of capital in order to develop their clinical programs. The capital-raising environment is always changing and there is risk that necessary capital to complete development may not be readily available. Furthermore, Maplight's pipeline consists of clinical-stage and pre-clinical programs, and its valuation will depend on the ability of these to continually advance.

ADDENDUM

Stocks Mentioned in Important Disclosures

Ticker	Company Name
AXSM	Axsome Therapeutics
TRDA	Entrada Therapeutics
MPLT	Maplight Therapeutics
UTHR	United Therapeutics

Analyst Certification

Each author of this research report hereby certifies that (i) the views expressed in the research report accurately reflect his or her personal views about any and all of the subject securities or issuers, and (ii) no part of his or her compensation was, is, or will be related, directly or indirectly, to the specific recommendations or views expressed in this report.

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Entrada Therapeutics is or has been in the past 12 months a client of TD Securities Inc., TD Securities (USA) LLC or its affiliates; TD Securities Inc., TD Securities (USA) LLC or its affiliates has provided investment banking services during the past 12 months.

TD Securities Inc., TD Securities (USA) LLC or its affiliates managed or co-managed a public offering of Entrada Therapeutics in the past 12 months.

TD Securities Inc., TD Securities (USA) LLC or its affiliates received compensation for investment banking services from Entrada Therapeutics in the past 12 months.

TD Securities Inc., TD Securities (USA) LLC or its affiliates expect to receive, or intend to seek, compensation for investment banking services in the next 3 months from Entrada Therapeutics.

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Buy (1): The stock is expected to achieve total return of +15% or more over the next 12 months

Hold (2): The stock is expected to achieve a total return that falls between -10% to +15% over the next 12 months

Sell (3): The stock is expected to achieve a total return of -10% or below over the next 12 months

Suspended (4): Due to evolving circumstances and potential conflicts of interest, we can no longer generate what we consider a defensible rating at the current time

Assumption: The expected total return calculation includes annual return and forecasted dividend yield

TD Cowen Equity Research Rating Distribution

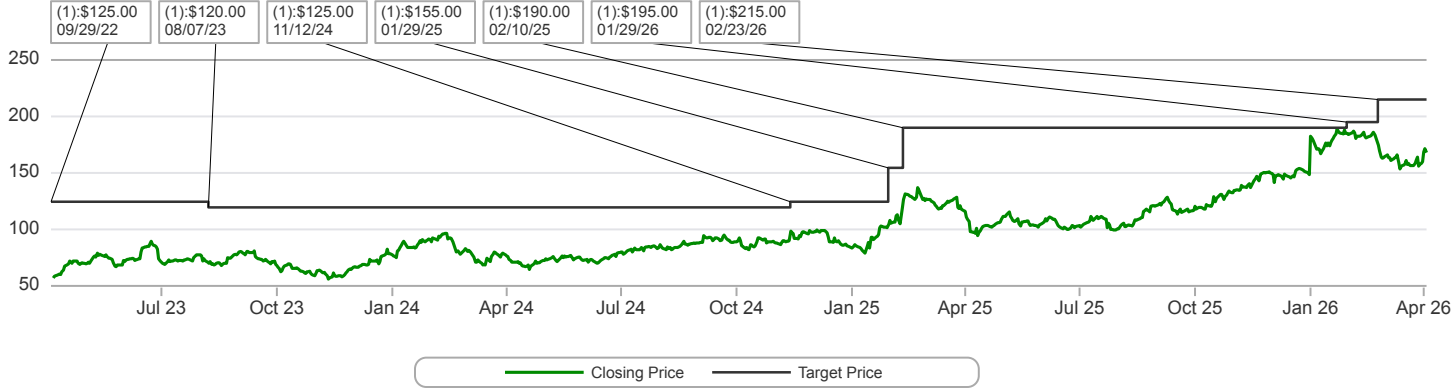
Distribution of Ratings/Investment Banking Services (IB) as of 03/31/26

Rating	Count	Ratings Distribution	Count	IB Services/Past 12 Months
Buy	848	65.63%	341	40.21%
Hold	429	33.20%	125	29.14%
Sell	15	1.16%	3	20.00%

TD Cowen Equity Research Rating Distribution Table does not include any subject company for which the equity research rating is currently suspended.

Axsome Therapeutics Rating History as of 04/02/2026

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Maplight Therapeutics Rating History as of 04/02/2026

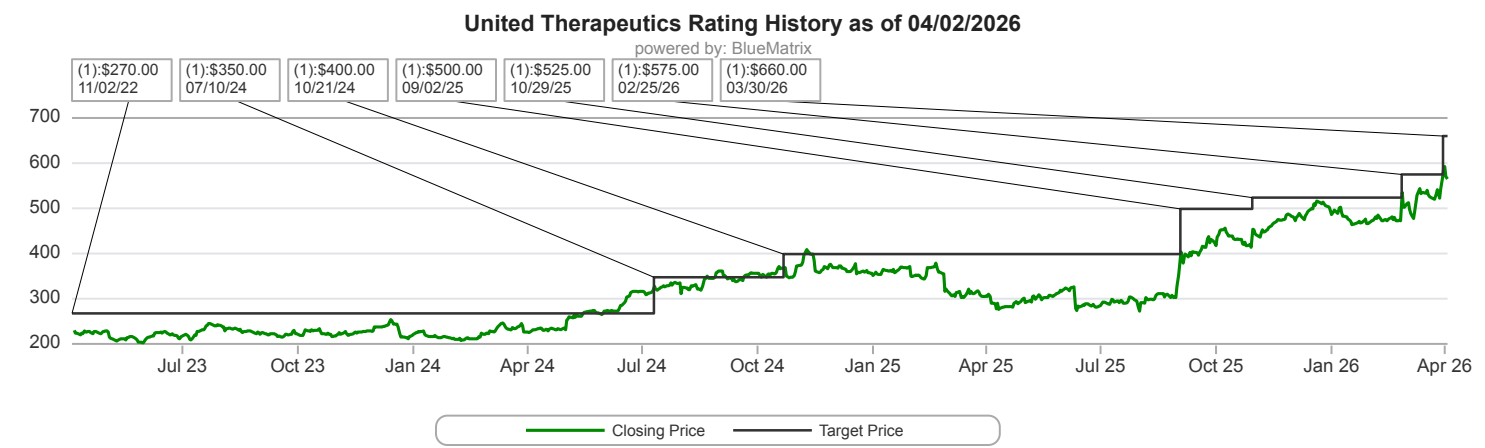
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Entrada Therapeutics Rating History as of 04/02/2026

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Legend for Price Chart:
I = Initiation | 1 = Buy | 2 = Hold | 3 = Sell | 4 = Suspended | UR = Price Target Under Review | D = Discontinued Coverage | \$xx = Price Target | NA = Not Available

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